MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, POWER, DIGITAL, KEY CODE CDB

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NAND gate module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Commmand, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 3.26 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be CDB.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 72.6 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/120-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

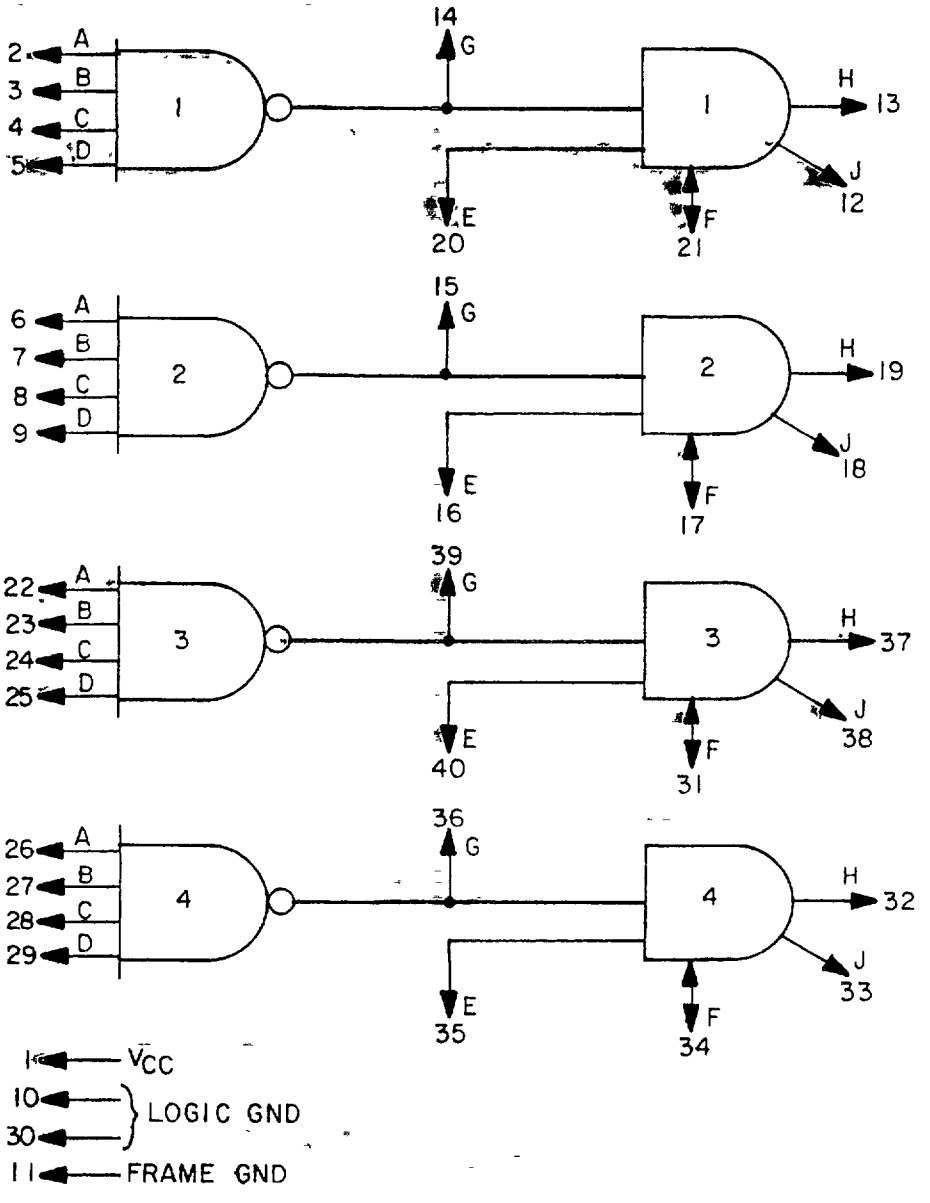
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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* FIGURE 1. CDB module function diagram, power NAND gate.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A4	Zener applied potential, condition A	III	S	0	400	0	450	0	415	0	465	mV
A5-A20	Zener applied potential, condition B	III	S	3.8	5.5	3.5	5 5	3.8	5 5	3.5	5.5	V
A21-24	Zener applied potential, condition C	III	S	0	400	0	450	0	415	0	465	mV
A25-A28	Zener applied potential, condition D	III	S	3 8	5 5	3.5	5 5	3.8	5.5	3.5	5.5	V
B1-B20	Reverse voltage	IV	S	0	400	0	450	0	415	0	465	mV
C1-C4	Logic output loaded	V	S	0.0	1 1	0.0	1 1	0.0	1.1	0.0	1.1	V
D1-D20	Static input current	VI	S	0	10	0	20	0	15	0	25	µA
E1	Isolation	VII	S	10	∞	10	∞	10	∞	10	∞	MΩ
E2	Continuity	VII	S	0	1	0	1	0	1	0	1	Ω
F1-F20	Input current, low level	VIII	X	0.0	1.6	0.0	1 6	0 0	1.6	0.0	1.6	mA
G1-G4	Input current (expander), low level	VIII	X	0.0	1.9	0.0	1.9	0.0	1.9	0.0	1.9	mA
H1-H52	Input current, high level	VIII	X	0	2	0	4	0	3	0	5	µA
J1-J4	Logic output short circuit current	VIII	X	0.70	1 31	0.70	1.31	0.70	1.31	0.70	1.31	mA
K1-K16	Output voltage, low level	IX	X	0	400	0	450	0	415	0	465	mV
L1-L16	Output voltage, high level	IX	X	3 80	5.05	3.50	5.05	3.80	5 05	3.50	5.05	V
M1-M8	Output voltage (driver), low level	IX	X	0	430	0	480	0	450	0	500	mV
N1-N8	Output voltage (driver), high level	IX	X	4.94	5.06	4.94	5.06	4.94	5.06	4.94	5.06	V
P1-P4	Expander input voltage	IX	X	0	450	0	480	---	---	---	---	mV
Q1-Q8	Driver output current, high level	X	X	0	5	0	50	0	200	0	250	µA

See footnotes at end of table.

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* TABLE I. Module electrical requirements -Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
R1	Power supply output current, low level	XI	S	0	120	0	120	0	120	0	120	mA
R2	Power supply output current, high level	XI	S	0	80	0	80	0	80	0	80	mA
S1-S20	Input capacitance	XII	Q	2	16	2	16	2	16	2	16	pF
T	Driver output isolation	XIII	Q	0	700	0	700	0	700	0	700	mV
U1-U4	Logic output crosstalk, low level	XIV	Q	0	800	0	800	0	800	0	800	mV
U5-U8	Logic output crosstalk, high level	XIV	Q	0	800	0	800	0	800	0	800	mV
V1-V4	Driver output crosstalk, low level	XIV	Q	0	400	0	400	0	400	0	400	mV
V5-V8	Driver output crosstalk, high level	XIV	Q	0	600	0	600	0	600	0	600	mV
W	Power supply filter capacitance	XV	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
X1-X4	Logic output voltage, high level	XVI	S	3.8	5.5	3.8	5.5	3.8	5.5	3.8	5.5	V
X5-X8	Logic output voltage, low level	XVI	S	0	400	0	450	0	420	0	470	mV
Y1-Y4	Transition time (t_{THL})	XVI	S	10	40	10	40	10	40	10	40	ns
Y5-Y8	Transition time (t_{TLH})	XVI	S	110	450	110	450	110	450	110	450	ns
Z1-Z4	Propagation delay time (t_{PLH})	XVI	S	25	100	25	100	25	100	25	100	ns
Z5-Z8	Propagation, delay time (t_{PHL})	XVI	S	3	75	3	75	3	75	3	75	ns
AA1-AA4	Driver output (high level), condition 1	XVII	S	4.5	5.5	4.5	5.5	4.5	5.5	4.5	5.5	V
AA5-AA8	Driver output (low level), condition 1	XVII	S	0	430	0	450	0	480	0	500	mV
BB1-BB4	Driver output transition time (t_{THL}), condition 1	XVII	S	8	100	8	100	8	100	8	100	ns
BB5-BB8	Driver output transition time (t_{TLH}), condition 1	XVII	S	8	150	8	150	8	175	8	175	ns
CC1-CC4	Driver output propagation delay time (t_{PLH}), condition 1	XVII	S	40	275	40	275	40	300	40	300	ns
CC5-CC8	Driver output propagation delay time (t_{PHL}), condition 1	XVII	S	20	175	20	175	20	200	20	200	ns

See footnotes at end of table.

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* TABLE I Module electrical requirements -Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
DD1-DD4	Driver output (high level), condition 2	XVIII	S	36	44	36	44	36	44	36	44	V
DD5-DD8	Driver output (low level), condition 2	XVIII	S	0	450	0	450	0	480	0	500	mV
EE1-EE4	Driver output transition time (t_{THL}), condition 2	XVIII	S	15	130	15	130	15	200	15	200	ns
EE5-EE8	Driver output transition time (t_{TLH}), condition 2	XVIII	S	1.0	3.0	1.0	3.0	1.0	3.0	1.0	3.0	µs
FF1-FF4	Driver output propagation delay time (t_{PLH}), condition 2	XVIII	S	200	900	200	900	200	1,000	200	1,000	ns
FF5-FF8	Driver output propagation delay time (t_{PHL}), condition 2	XVIII	S	30	200	30	200	30	300	30	300	ns

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

4.4.2 Order of testing. Test conditions A through FF shall be performed in any order with the following exceptions. If test conditions A, B, C, or Q are part of a sequence of tests, the order of testing shall be as follows: A, B, C, Q where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through FF in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions F through N, Q, R, and T through FF shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 6. Following the life test and while still at the maximum class temperature, electrical tests F through N, Q, R, and T through FF shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests F through FF shall be performed in the order specified in 4.4.2 using the 25°±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Military Specification MIL-M-28787/120 supersedes Department of the Navy, Naval Ordance Systems Command Drawing No. 10001 - 2658712. Part number M28787/120-1 supersedes part numbers 2658712 and M28787/120.

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Power dissipation	
All outputs at low level	1.1 watts (outputs loaded)
All outputs at high level	0.45 watt (outputs loaded)
Supply voltage	5 Vdc ±10 percent

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Logic input voltage	
Low level	0.0 V minimum, 1.2 V maximum
High level	1.8 V minimum, 5.5 V maximum
Logic input current	
High level (V_{IN} = 4 V)	5 µA maximum
(V_{IN} = 5 V)	25 µA maximum
Low level (V_{IN} = 0 V)	-1.6 mA maximum
Expander input voltage	
High level	1.8 V minimum, 5.5 V maximum
Expander input current	
Low level (V_{IN} = 0 V)	-1.9 mA maximum
Logic output voltage	
Low level	0.0 V minimum, 0.46 V maximum
High level	3.5 V minimum, 5.5 V maximum
Driver output voltage	
Low level	0.0 V minimum, 0.50 V maximum
High level	4.94 V minimum, 5.06 V maximum
Driver output current	
High level (V_{OUT} = 40 V)	250 µA maximum
Driver output drive current	
Low level	200 mA maximum
Speed	DC to 3 MHz

* 6.5 Module function. The module consists of four independent circuits satisfying the following logic equations:

$$(A \cdot B \cdot C \cdot D) = \overline{G}$$
$$(A \cdot B \cdot C \cdot D) + E + F = \overline{H}$$

* 6.5.1 Input to output relationship. The input to output relationship is specified in figure 2 and table II.

* 6.5.1.1 Expander input. Expander input F is normally used with a series diode external to the module. A high level is applied to F only through the diode (anode wired to F).

* 6.5.2 Loading. Each logic output (G) is capable of driving six standard logic loads (9.6 mA) at a low level or eleven standard logic loads (55 µA) at a high level. The driver output (H) is capable of driving a load with a maximum ON current of 200 mA and a maximum OFF voltage of +40.0 volts.

* 6.5.3 Parallel loading. Paralleling of the logic output (G) with the output of a standard NAND gate is permissible with a reduction in low level logic fan-out equal to 5/6 standard logic loads times the number of standard NAND gates in parallel with the logic output.

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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TABLE II. Input to output relationship. 1/-4/

Inputs					Expander input	Outputs	
A	B	C	D	E	F	G	H
L	X	X	X	X	Y	H	X
X	L	X	X	X	Y	H	X
X	X	L	X	X	Y	H	X
X	X	X	L	X	Y	H	X
X	X	X	X	L	Y	X	L
X	X	X	X	X	L	X	L
H	H	H	H	X	Y	L	L

1/ X = Signifies either a logic low level or a logic high level
2/ Y = Signifies either a logic low level or an open input.
3/ L = Low level.
4/ H = High level.

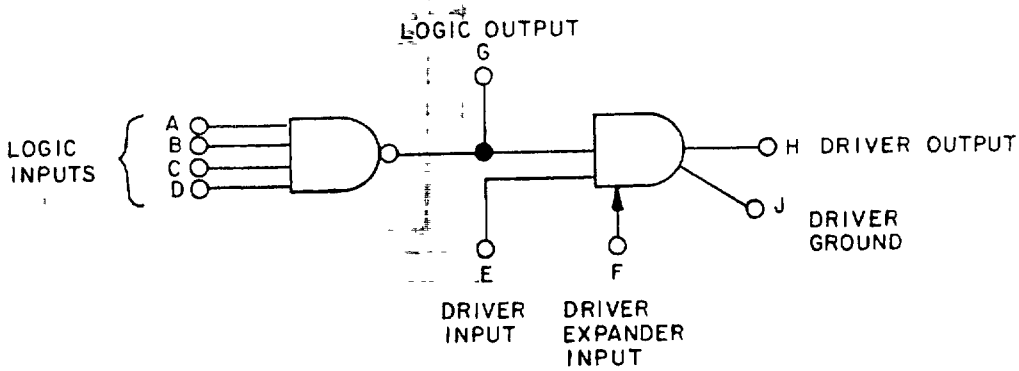
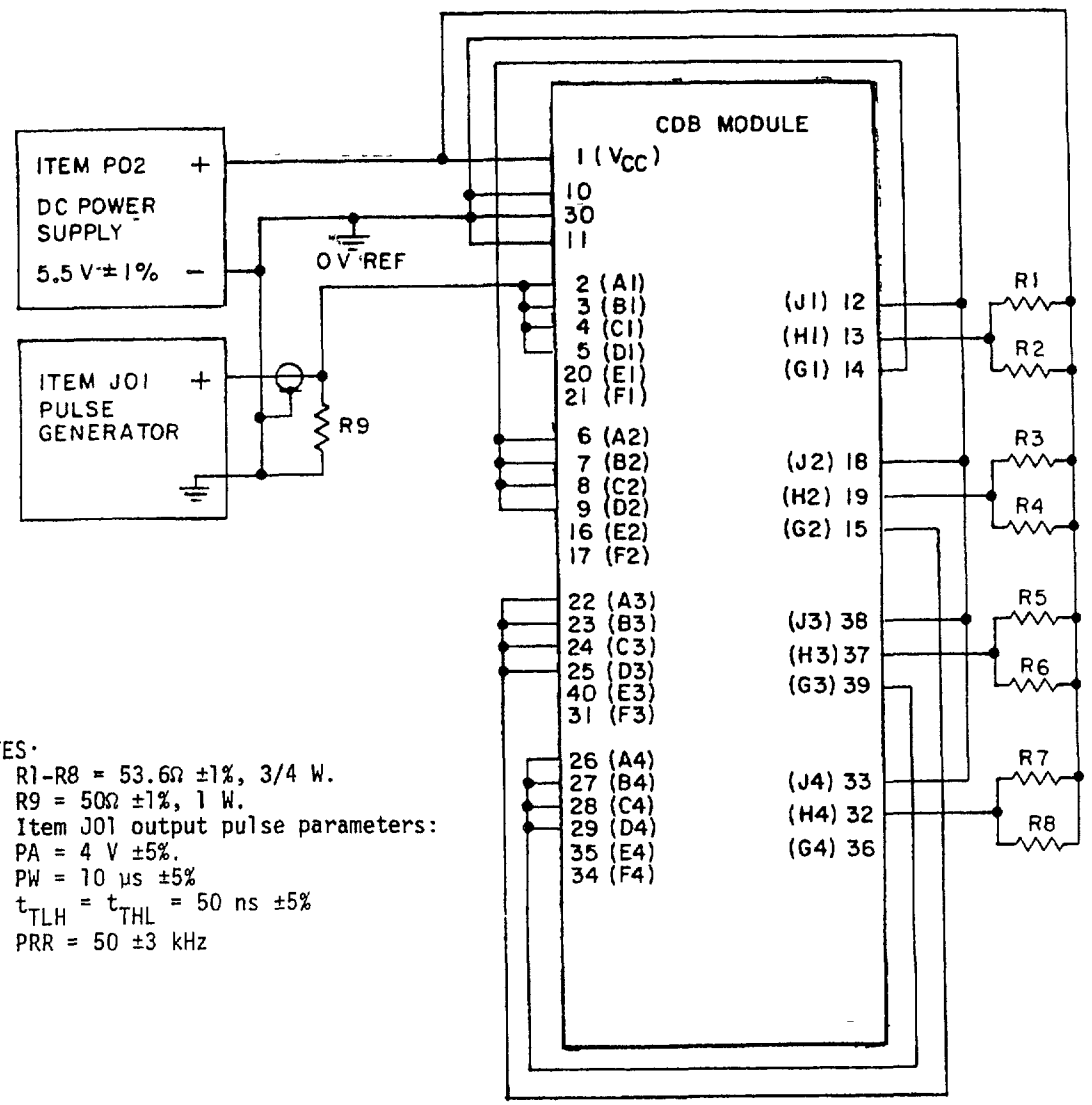


FIGURE 2. Logic input to output.

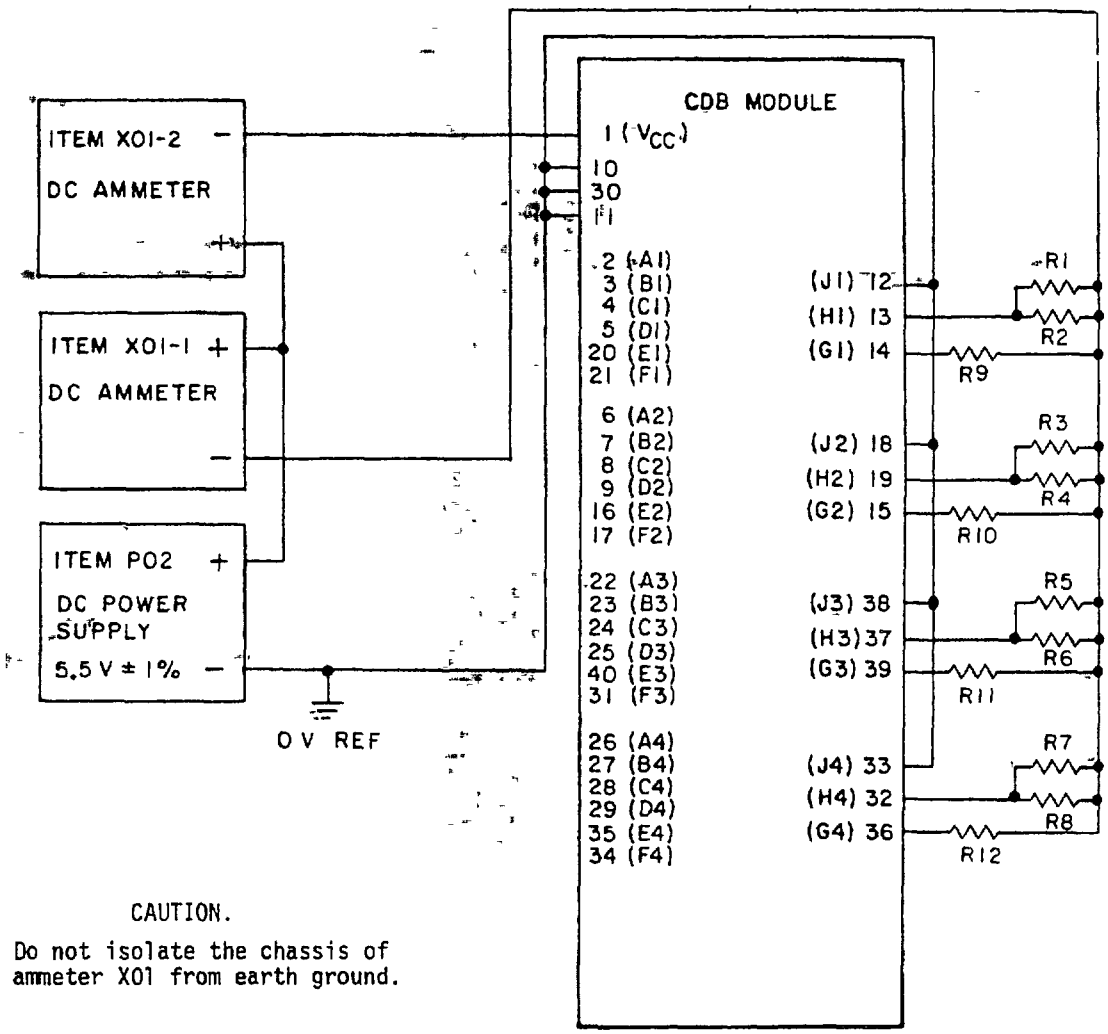
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NOTES:
1. R1-R8 = 53.6Ω ±1%, 3/4 W.
2. R9 = 50Ω ±1%, 1 W.
3. Item J01 output pulse parameters:
   PA = 4 V ±5%.
   PW = 10 µs ±5%
   $t_{TLH} = t_{THL}$ = 50 ns ±5%
   PRR = 50 ±3 kHz

FIGURE 3. Operating temperature test circuit

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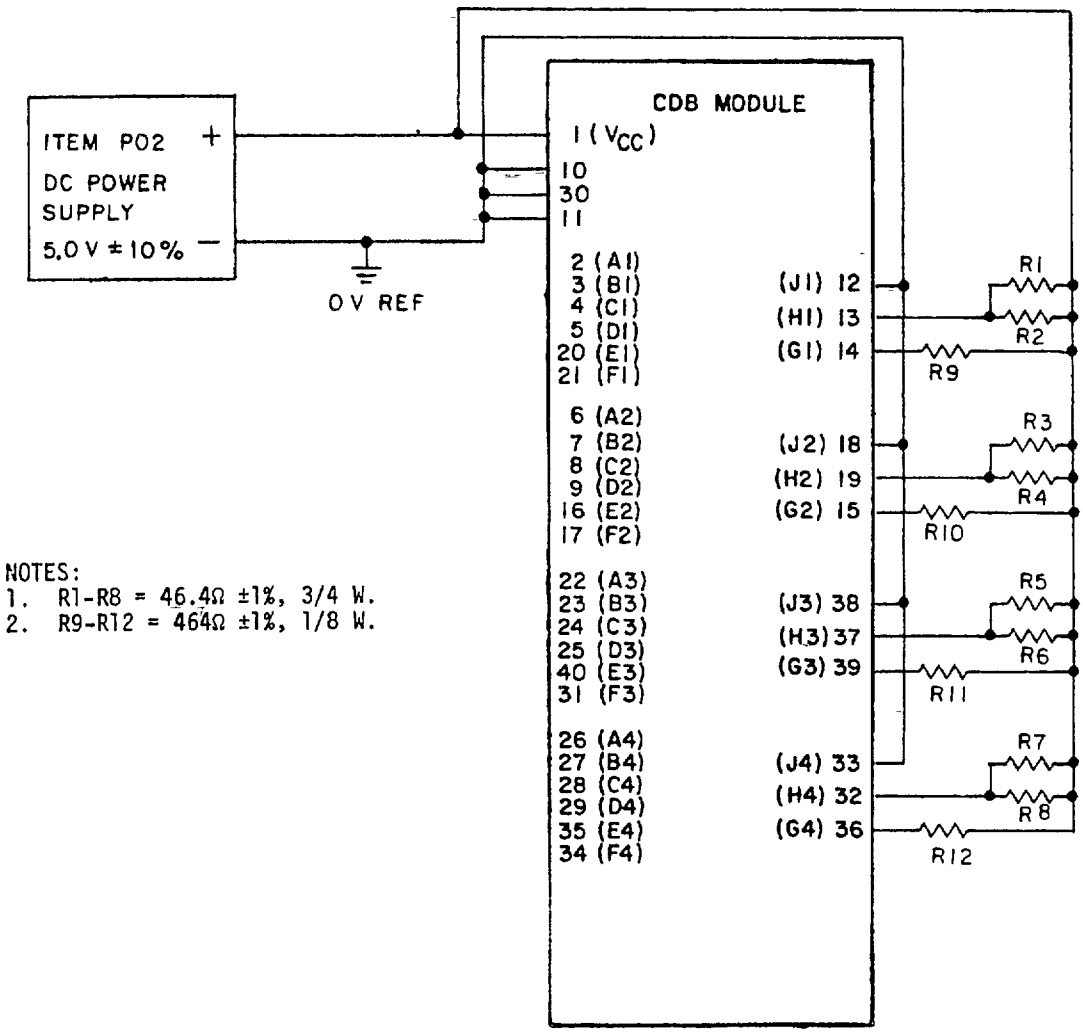
CAUTION.

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES:
1. R1-R8 = 53.6Ω ±1%, 3/4 W.
2. R9-R12 = 464Ω ±1%, 1/8 W.

FIGURE 4. Component temperature test circuit.

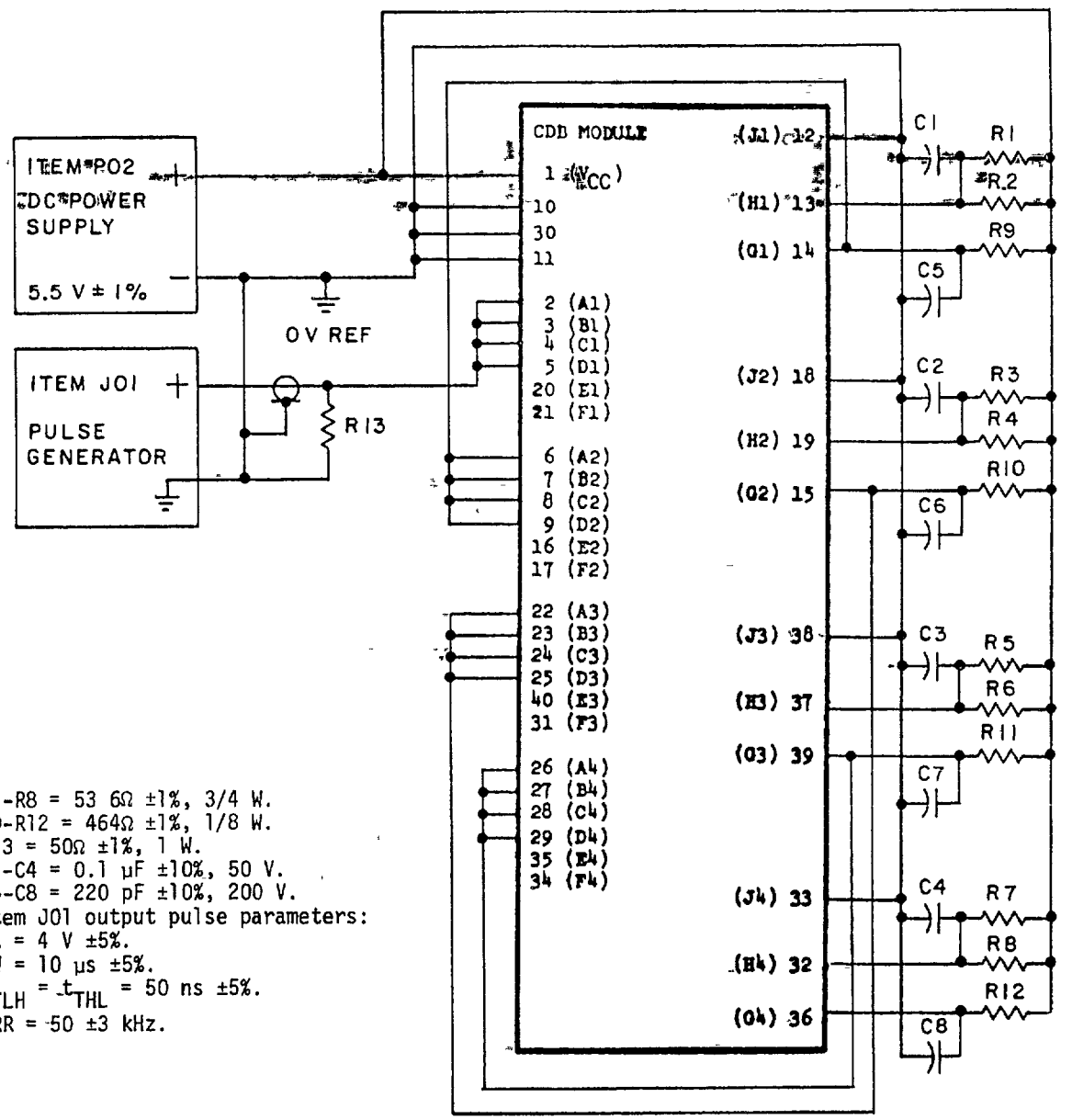
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NOTES:
1. R1-R8 = 46.4Ω ±1%, 3/4 W.
2. R9-R12 = 464Ω ±1%, 1/8 W.

* FIGURE 5. Vibration test circuit.

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NOTES
1. R1-R8 = 53.6Ω ±1%, 3/4 W.
2. R9-R12 = 464Ω ±1%, 1/8 W.
3. R13 = 50Ω ±1%, 1 W.
4. C1-C4 = 0.1 µF ±10%, 50 V.
5. C5-C8 = 220 pF ±10%, 200 V.
6. Item J01 output pulse parameters:
   PA = 4 V ±5%.
   PW = 10 µs ±5%.
   $t_{TLH} = t_{THL}$ = 50 ns ±5%.
   PRR = 50 ±3 kHz.

FIGURE 6. Life test circuit.

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* TABLE III. Zener applied potential (conditions A, B, C, and D) test.

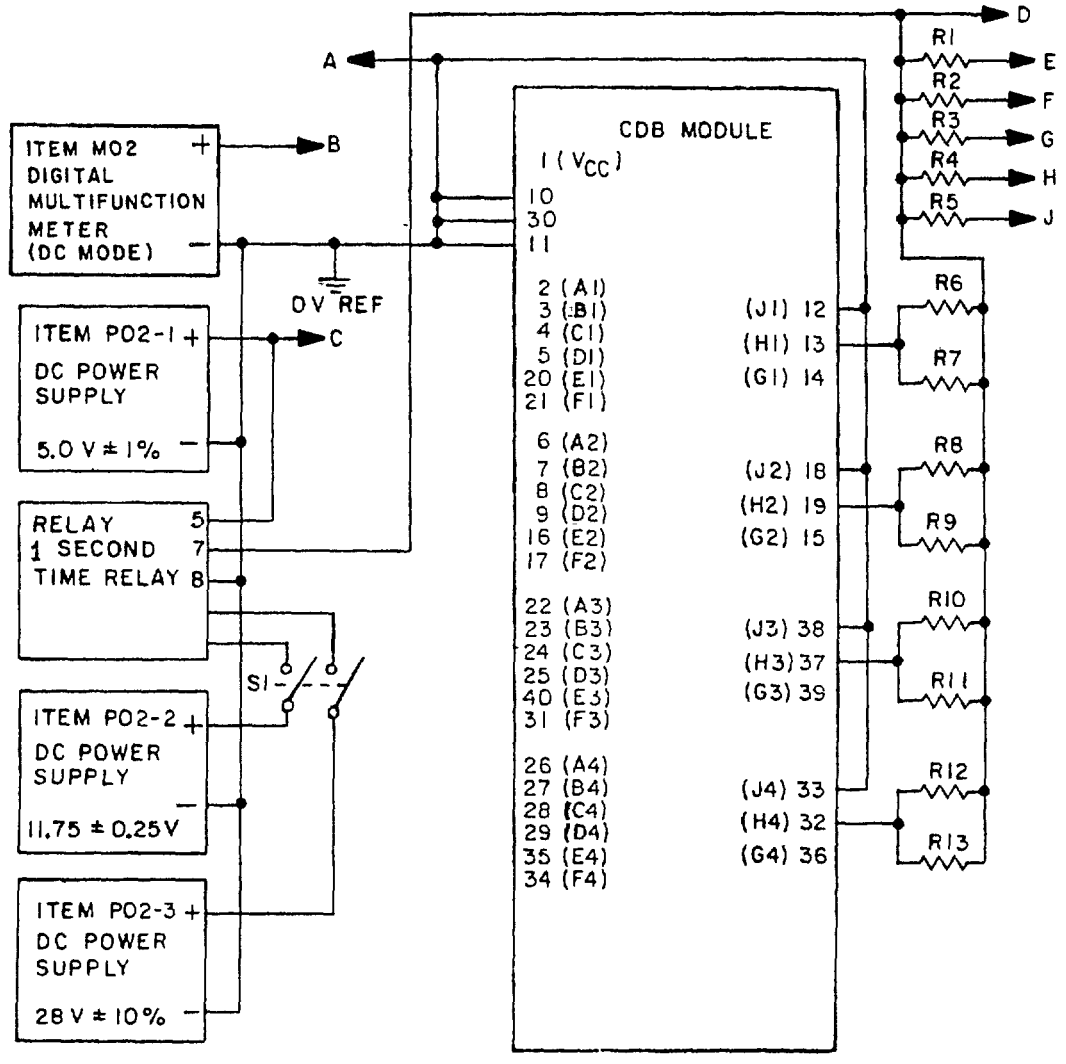
Test condition	Parameter		Connect									Initial limits 25 ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	J to pin	Min	Max		
A1	G1	Zener applied potential, condition A	Open	14	Open	1	20	2	3	4	5	0	400	mV	7
A2	G2			15			16	6	7	8	9				
A3	G3			39			40	22	23	24	25				
A4	G4			36			35	26	27	28	29				
A5	G1	Zener applied potential, condition B	3,4,5	14			2	20	Open	Open	Open	3.8	5.5	V	
A6	G1		2,4,5				3	Open							
A7	G1		2,3,5				4	Open							
A8	G1		2,3,4				5	Open							
A9	G2		7,8,9	15			6	16							
A10	G2		6,8,9				7	Open							
A11	G2		6,7,9				8	Open							
A12	G2		6,7,8				9	Open							
A13	G3		23,24, 25	39			22	40							
A14	G3		22,24, 25				23	Open							
A15	G3		22,23, 25				24	Open							
A16	G3		22,23, 24				25	Open							
A17	G4		27,28, 29	36			26	35							
A18	G4		26,28. 29	36			27	Open							

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* TABLE III. Zener applied potential (conditions A, B, C, and D) test -Continued.

Test condition	Parameter	Connect									Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	J to pin	Min	Max		
A19	G4 Zener applied potential, condition B	26,27, 29	36	Open	1	28	Open	Open	Open	Open	3.8	5.5	V	7
A20	G4 Zener applied potential, condition B	26,27, 28	36	Open	1	29	↓	↓	↓	↓	3.8	5.5	V	↓
A21	G1 Zener applied potential, condition C	20	14	1	Open	2	↓	↓	↓	↓	0	400	mV	↓
A22	G2 ↓	16	15	↓	↓	6	↓	↓	↓	↓	↓	↓	↓	↓
A23	G3 ↓	40	39	↓	↓	22	↓	↓	↓	↓	↓	↓	↓	↓
A24	G4 ↓	35	36	↓	↓	26	↓	↓	↓	↓	↓	↓	↓	↓
A25	G1 Zener applied potential, condition D	2	14	↓	↓	20	↓	↓	↓	↓	3.8	5.5	V	↓
A26	G2 ↓	6	15	↓	↓	16	↓	↓	↓	↓	↓	↓	↓	↓
A27	G3 ↓	22	39	↓	↓	40	↓	↓	↓	↓	↓	↓	↓	↓
A28	G4 ↓	26	36	↓	↓	35	↓	↓	↓	↓	↓	↓	↓	↓

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NOTES:
1. R1 = 154Ω ±1%, 1/2 W.
2. R2-R5 = 5.9 kΩ ±1%, 1/8 W.
3. R6-R13 = 46.4Ω ±1%, 3/4 W.
4. Relay = Agastat Model 2112, or equivalent. The relay shall be adjusted for a 1 second (±20%) delay.
5. S1 = DPST switch.
6. For each test condition, switch S1 shall be closed and voltage readings will be made after 1.2 seconds. At the conclusion of each test condition, switch S1 will be opened.

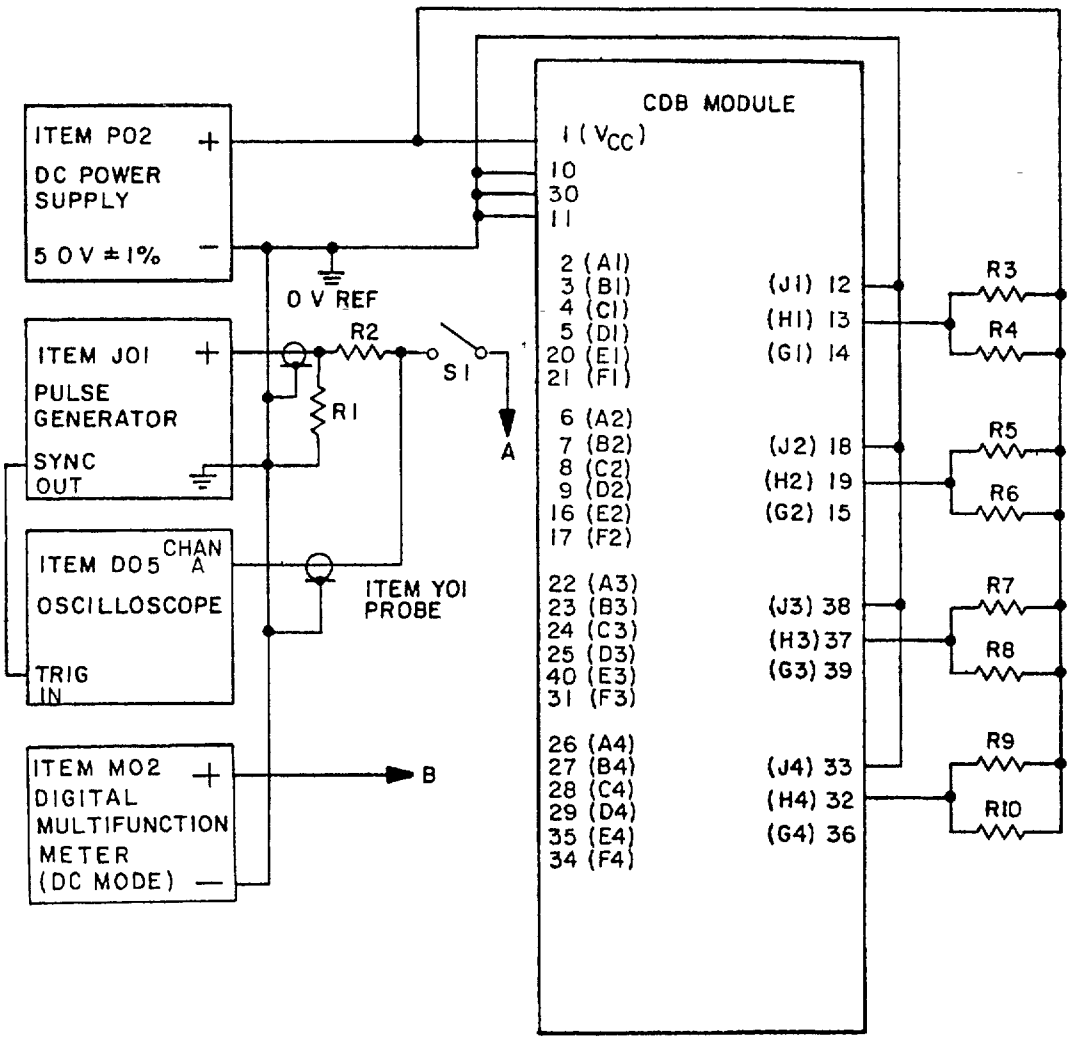
* FIGURE 7. Zener applied potential (conditions A,B,C, and D) test circuit for table III.

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* TABLE IV. Reverse voltage test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
B1	G1 Reverse voltage	2	14	0	400	mV	8,9
B2	G1	3	14				
B3	G1	4	14				
B4	G1	5	14				
B5	H1	20	13				
B6	G2	6	15				
B7	G2	7	15				
B8	G2	8	15				
B9	G2	9	15				
B10	H2	16	19				
B11	G3	22	39				
B12	G3	23	39				
B13	G3	24	39				
B14	G3	25	39				
B15	H3	40	37				
B16	G4	26	36				
B17	G4	27	36				
B18	G4	28	36				
B19	G4	29	36				
B20	H4	35	32				

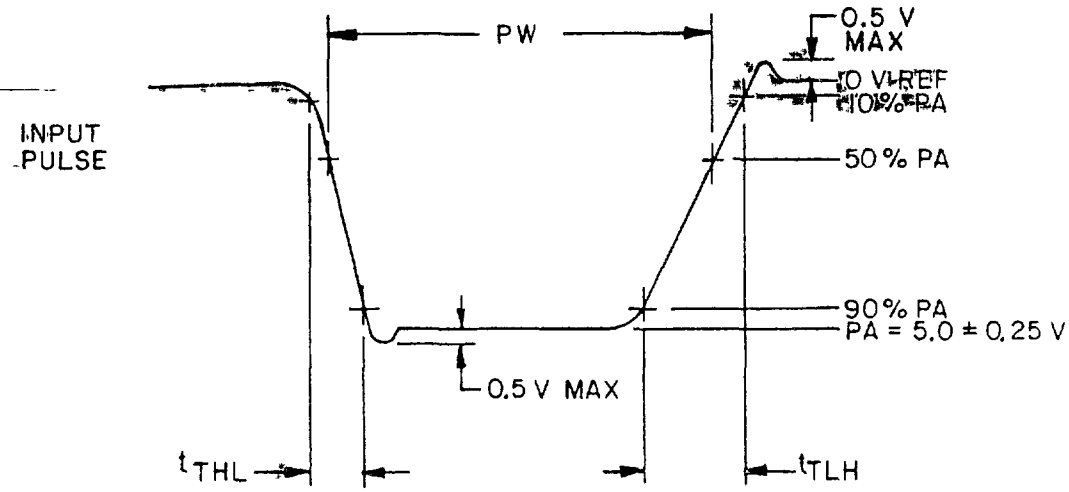
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NOTES:
1. R1 = 50Ω ±1%, 1 W.
2. R2 = 121Ω ±1%, 1/8 W.
3. R3-R10 = 46.4Ω ±1%, 1/8 W.
4. S1 = SPST switch.
5. Connect A to pins designated in table. Close switch S1 for a minimum of 10 seconds.
6. Open switch S1 and connect B to pins designated in table. The module shall conform to the requirements of the table.

* FIGURE 8. Reverse voltage test circuit for table IV.

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NOTE: Input pulse parameters
PW = 10 ±1 µs.
$t_{TLH} = t_{THL}$ = 50 ±10 ns.
PRR = 30 kHz ±10%.

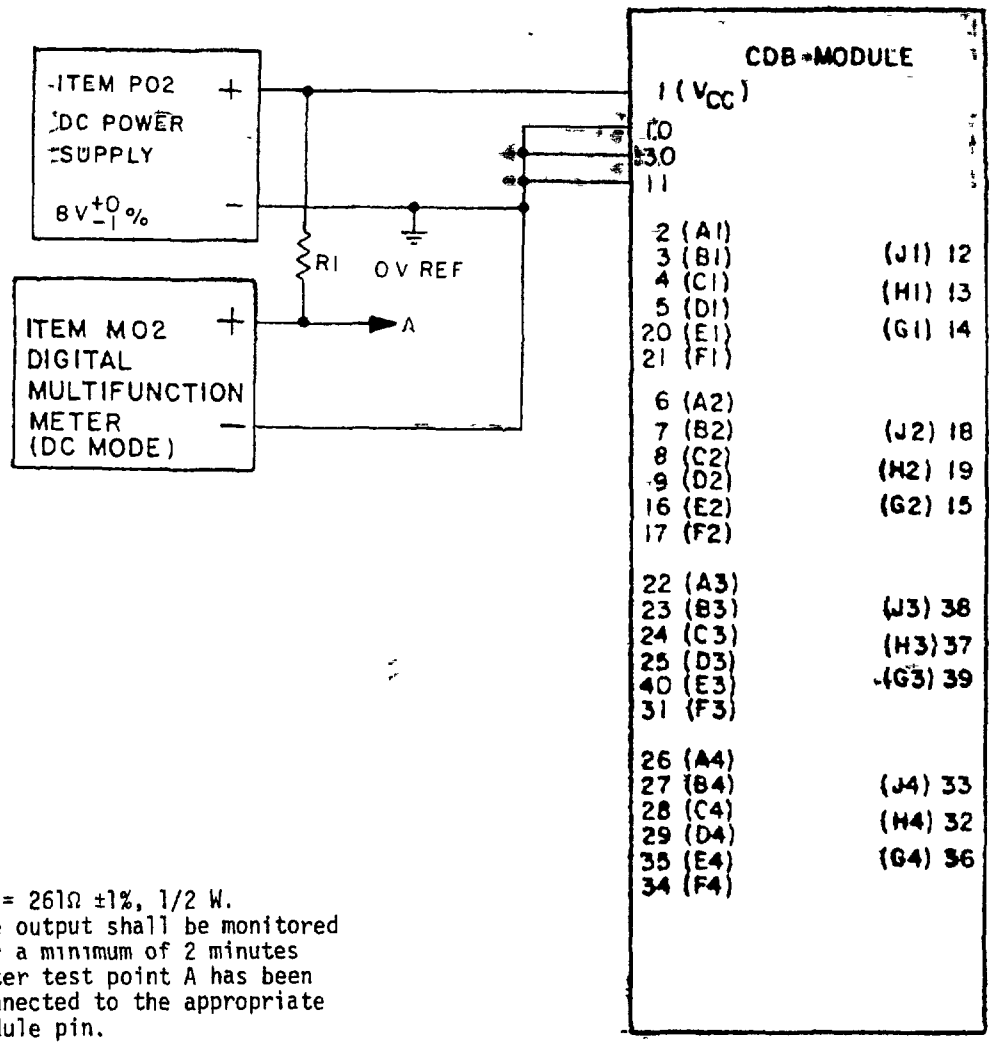
FIGURE 9. Reverse voltage test circuit waveform for table IV.

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TABLE V. Logic output loaded test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C1	G1 Logic output loaded	14	0.0	1.1	V	10
C2	G2	15				
C3	G3	39				
C4	G4	36				

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NOTES:
1. R1 = 261Ω ±1%, 1/2 W.
2. The output shall be monitored for a minimum of 2 minutes after test point A has been connected to the appropriate module pin.

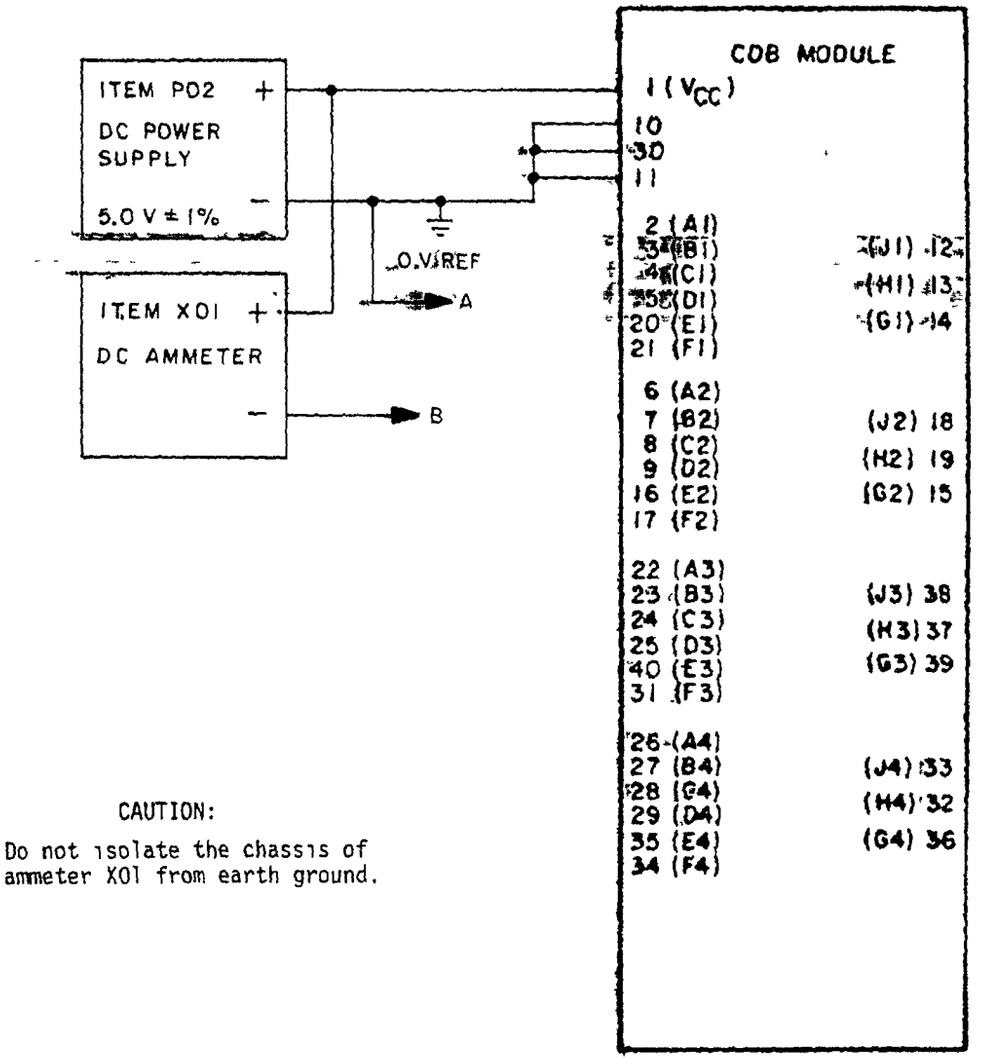
* FIGURE 10 Logic output loaded test circuit for table V

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TABLE VI. Static input current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max[1]		
D1	A1 Static input current	3,4,5	2	0	10	µA	11
D2	B1	2,4,5	3				
D3	C1	2,3,5	4				
D4	D1	2,3,4	5				
D5	A2	7,8,9	6				
D6	B2	6,8,9	7				
D7	C2	6,7,9	8				
D8	D2	6,7,8	9				
D9	A3	23,24,25	22				
D10	B3	22,24,25	23				
D11	C3	22,23,25	24				
D12	D3	22,23,24	25				
D13	A4	27,28,29	26				
D14	B4	26,28,29	27				
D15	C4	26,27,29	28				
D16	D4	26,27,28	29				
D17	E1	14	20				
D18	E2	15	16				
D19	E3	39	40				
D20	E4	36	35				

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

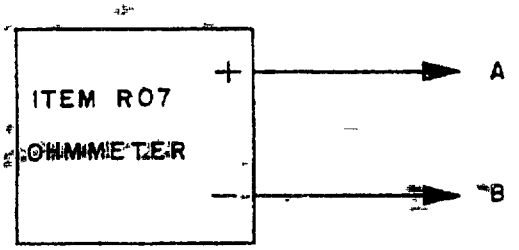
FIGURE 11. Static input current test circuit for table VI.

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* TABLE VII. Isolation and continuity test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	Isolation	11	All other pins	10	∞	MΩ	12
E2	Continuity	11	Frame	0	1	Ω	12

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CDB MODULE

1 (V_{CC})	
10	
30	
11	
2 (A1)	
3 (B1)	(J1) 12
4 (C1)	(H1) 13
5 (D1)	(G1) 14
20 (E1)	
21 (F1)	
6 (A2)	
7 (B2)	(J2) 18
8 (C2)	(H2) 19
9 (D2)	(G2) 15
16 (E2)	
17 (F2)	
22 (A3)	
23 (B3)	(J3) 38
24 (C3)	(H3) 37
25 (D3)	(G3) 39
40 (E3)	
31 (F3)	
26 (A4)	
27 (B4)	(J4) 33
28 (C4)	(H4) 32
29 (D4)	(G4) 36
35 (E4)	
34 (F4)	

NOTE: For test condition E2 penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

* FIGURE 12. Isolation and continuity test circuit for table VII.

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TABLE VIII. Input current (low-level, high-level, and expander) and logic output short circuit current test.

Test condition		Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
F1	A1	Input current, low level	Open	Open	2	0.0	1.6	mA	13
F2	B1				3				
F3	C1				4				
F4	D1				5				
F5	A2				6				
F6	B2				7				
F7	C2				8				
F8	D2				9				
F9	A3				22				
F10	B3				23				
F11	C3				24				
F12	D3				25				
F13	A4				26				
F14	B4				27				
F15	C4				28				
F16	D4				29				
F17	E1			2	20				
F18	E2			6	16				
F19	E3			22	40				
F20	E4			26	35				
G1	F1	Input current (expander), low level		2	21		1.9		
G2	F2			6	17				
G3	F3			22	31				
G4	F4			26	34				

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TABLE VIII. Input current (low-level, high-level, and expander) and logic output short circuit current test -Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	A1 Input current, high level	2	3	Open	0	2	µA	13
H2	A1	2	4					
H3	A1	2	5					
H4	B1	3	2					
H5	B1	3	4					
H6	B1	3	5					
H7	C1	4	2					
H8	C1	4	3					
H9	C1	4	5					
H10	D1	5	2					
H11	D1	5	3					
H12	D1	5	4					
H13	A2	6	7					
H14	A2	6	8					
H15	A2	6	9					
H16	B2	7	6					
H17	B2	7	8					
H18	B2	7	9					
H19	C2	8	6					
H20	C2	8	7					
H21	C2	8	9					
H22	D2	9	6					
H23	D2	9	7					

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TABLE VIII. Input current (low-level, high-level, and expander) and logic output short circuit current test -Continued.

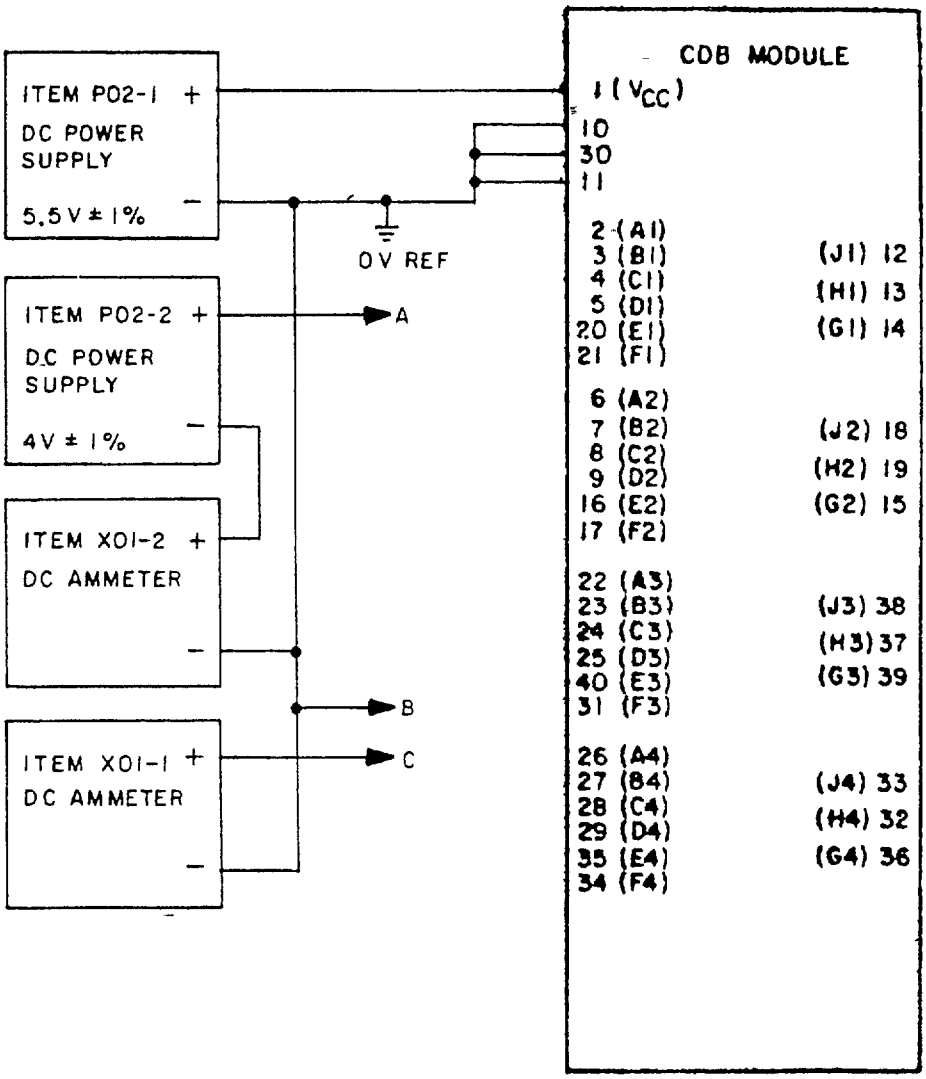
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H24	D2 Input current, high level	9	8	Open	0	2	µA	13
H25	A3	22	23					
H26	A3	22	24					
H27	A3	22	25					
H28	B3	23	22					
H29	B3	23	24					
H30	B3	23	25					
H31	C3	24	22					
H32	C3	24	23					
H33	C3	24	25					
H34	D3	25	22					
H35	D3	25	23					
H36	D3	25	24					
H37	A4	26	27					
H38	A4	26	28					
H39	A4	26	29					
H40	B4	27	26					
H41	B4	27	28					
H42	B4	27	29					
H43	C4	28	26					
H44	C4	28	27					
H45	C4	28	29					

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TABLE VIII. Input current (low-level, high-level, and expander) and logic output short circuit current test -Continued.

Test condition		Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
H46	D4	Input current, high level	29	26	Open	0	2	µA	13
H47	D4		29	27					
H48	D4		29	28					
H49	E1		20	14					
H50	E2		16	15					
H51	E3		40	39					
H52	E4		35	36					
J1	G1	Logic output short circuit current	Open	2,21	14	0.70	1.31	mA	
J2	G2			6,17	15				
J3	G3			22,31	39				
J4	G4			26,34	36				

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground

NOTE: Since items XO1-1 and XO1-2 are not used simultaneously, only one item XO1 is required

* FIGURE 13. Input current (low-level, high-level, and expander) and logic output short circuit test circuit for table VIII.

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* TABLE IX. Output voltage (low-level and high-level), driver (low-level and high-level), and expander input voltage test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
K1	G1 Output voltage, low level	14	Open	Open	2	Open	14	0	400	mV	14
K2	G1	14			3		14				
K3	G1	14			4		14				
K4	G1	14			5		14				
K5	G2	15			6		15				
K6	G2	15			7		15				
K7	G2	15			8		15				
K8	G2	15			9		15				
K9	G3	39			22		39				
K10	G3	39			23		39				
K11	G3	39			24		39				
K12	G3	39			25		39				
K13	G4	36			26		36				
K14	G4	36			27		36				
K15	G4	36			28		36				
K16	G4	36			29		36				
L1	G1 Output voltage, high level	Open			2	14	14	3.80	5.05	V	
L2	G1				3						
L3	G1				4						
L4	G1				5						
L5	G2				6	15	15				
L6	G2				7						
L7	G2				8						
L8	G2				9						

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* TABLE IX. Output voltage (low-level and high level), driver (low-level and high level), and expander input voltage test -Continued.

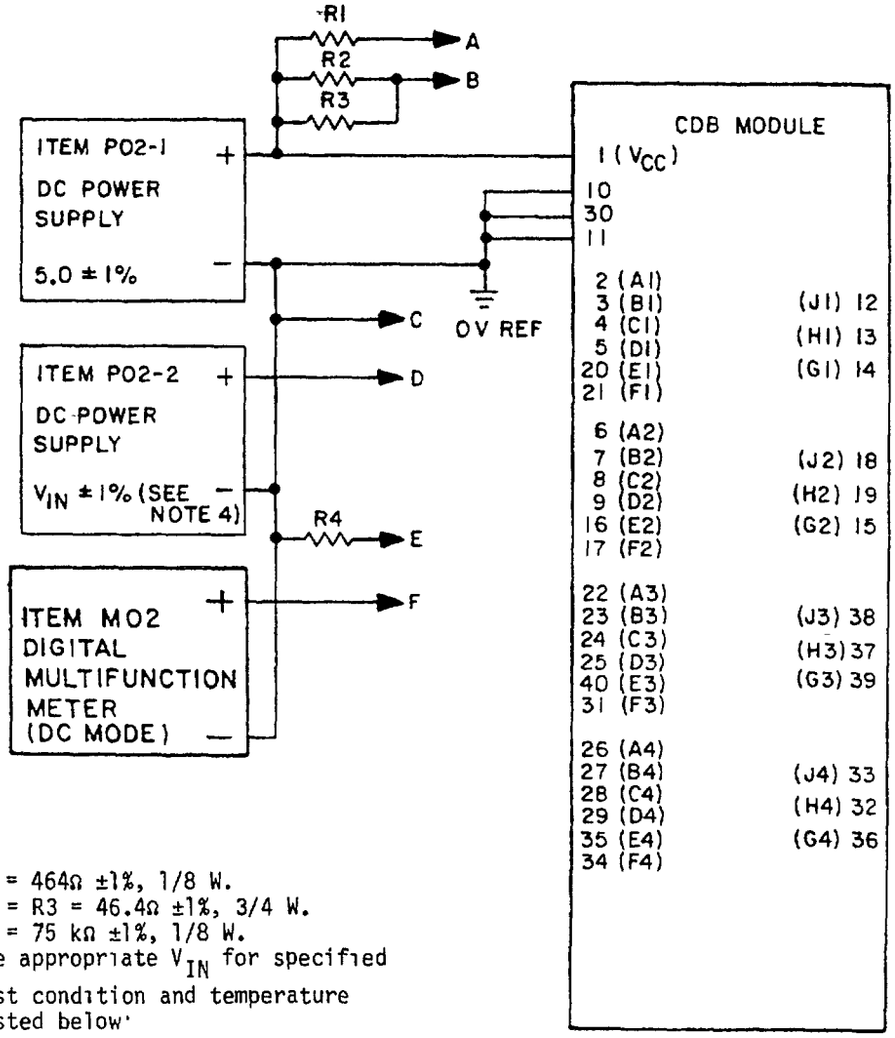
Test condition		Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
L9	G3	Output voltage, high level	Open	Open	Open	22	39	39	3.80	5.05	V	14
L10	G3					23						
L11	G3					24						
L12	G3					25						
L13	G4					26	36	36				
L14	G4					27						
L15	G4					28						
L16	G4					29						
M1	H1	Output voltage (driver), low level		13	2,12	20	Open	13	0	430	mV	
M2	H1			13	2,12	14		13				
M3	H2			19	6,18	16		19				
M4	H2			19	6,18	15		19				
M5	H3			37	22,38	40		37				
M6	H3			37	22,38	39		37				
M7	H4			32	26,33	35		32				
M8	H4			32	26,33	36		32				
N1	H1	Output voltage (driver), high level		13	2,12	20		13	4.94	5.06	V	
N2	H1			13	2,12	14		13				
N3	H2			19	6,18	16		19				
N4	H2			19	6,18	15		19				
N5	H3			37	22,38	40		37				
N6	H3			37	22,38	39		37				
N7	H4			32	26,33	35		32				
N8	H4			32	26,33	36		32				

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* TABLE IX. Output voltage (low-level and high-level), driver (low-level and high-level), and expander input voltage test -Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
P1	H1 Expander input voltage	Open	13	2,12	21	Open	13	0	450	mV	14
P2	H2	↓	19	6,18	17	↓	19	↓	↓	↓	↓
P3	H3	↓	37	22,38	31	↓	37	↓	↓	↓	↓
P4	H4	↓	32	26,33	34	↓	32	↓	↓	↓	↓

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NOTES:
1. R1 = 464Ω ±1%, 1/8 W.
2. R2 = R3 = 46.4Ω ±1%, 3/4 W.
3. R4 = 75 kΩ ±1%, 1/8 W.
4. Use appropriate V_{IN} for specified test condition and temperature listed below.

Test condition	Temperature (°C)	V_{IN}
K1-K16, N1-N8	0	2.0
	25	1.9
	60	1.8
	80	1.8
L1-L16, M1-M8	0	1.20
	25	1.10
	60	0.95
	80	0.80
P1-P4	25	1.8

0°, 60°, and 80°C refer to operating temperatures (module frame); 25°C refers to ambient temperature. For 80°C use the 0° to 60°C end-of-life limits.

* FIGURE 14. Output voltage (low-level and high-level), driver (low-level and high-level), and expander input voltage test circuit for table IX.

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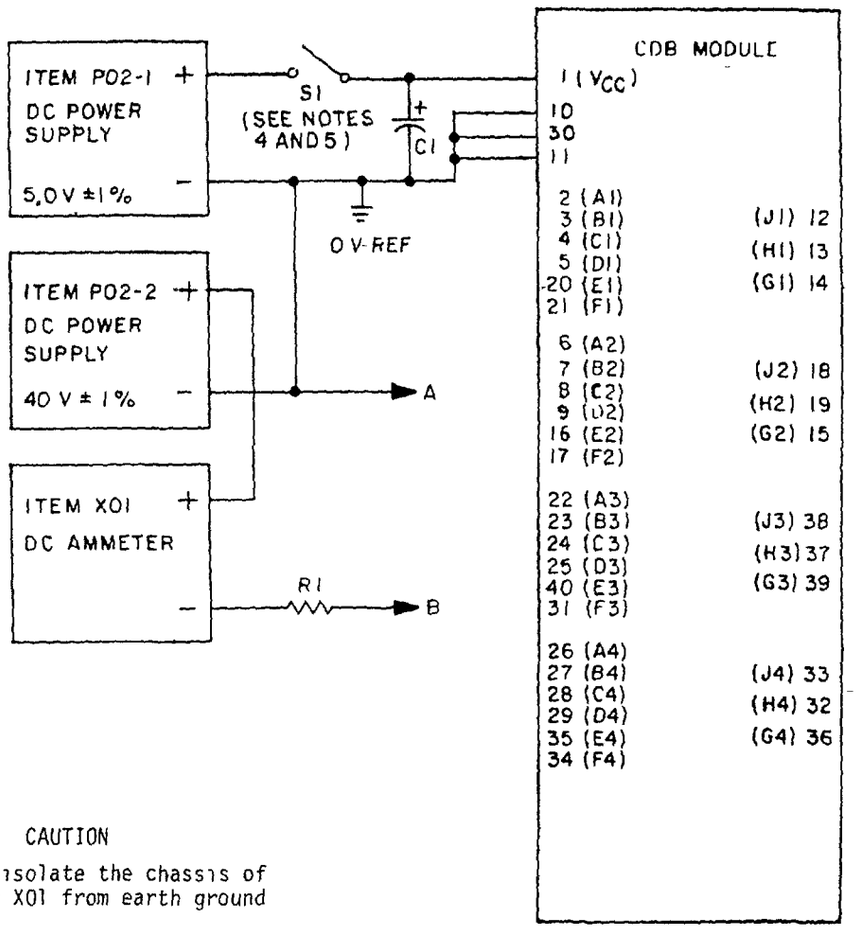
TABLE X. Driver output current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Q1	H1 Driver output current, high level	2,12	13	0	5	µA	15
Q2	H1	2,12	13				
Q3	H2	6,18	19				
Q4	H2	6,18	19				
Q5	H3	22,38	37				
Q6	H3	22,38	37				
Q7	H4	26,33	32				
Q8	H4	26,33	32				

TABLE XI. Power supply output current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
R1	Power supply output current, low level	1	10	0	120	mA	16
R2	Power suppply output current, high level	10	1	0	80	mA	16

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CAUTION

Do not isolate the chassis of ammeter XO1 from earth ground

NOTES:
1. R1 = 205Ω ±1%, 10 W.
2. C1 = 10 µF ±10%, 75 VDc.
3. S1 = SPST switch.
4. For test conditions Q1, Q3, Q5, and Q7 switch S1 is closed.
5. For test conditions Q2, Q4, Q6, and Q8 switch S1 is opened after points A and B are connected and power is applied.

*FIGURE 15. Driver output current (high-level) test circuit for table X.

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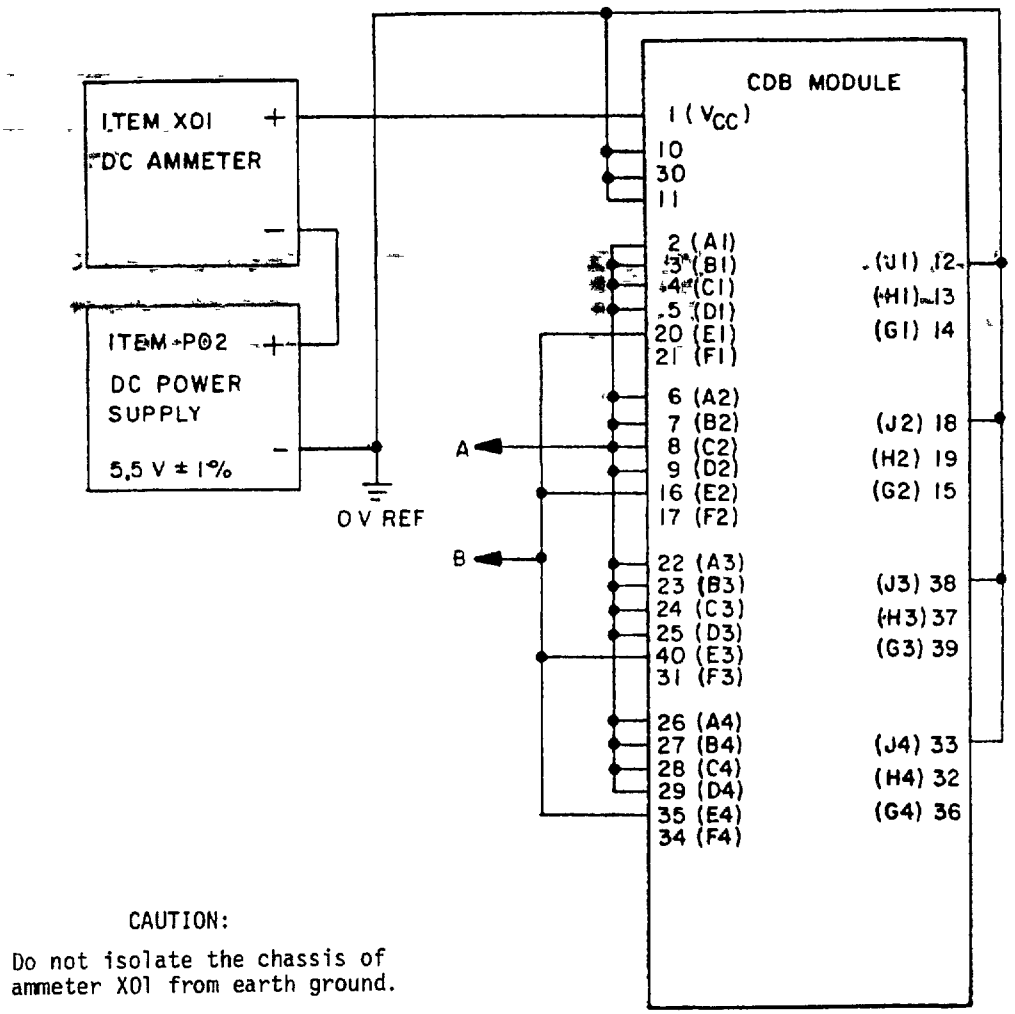


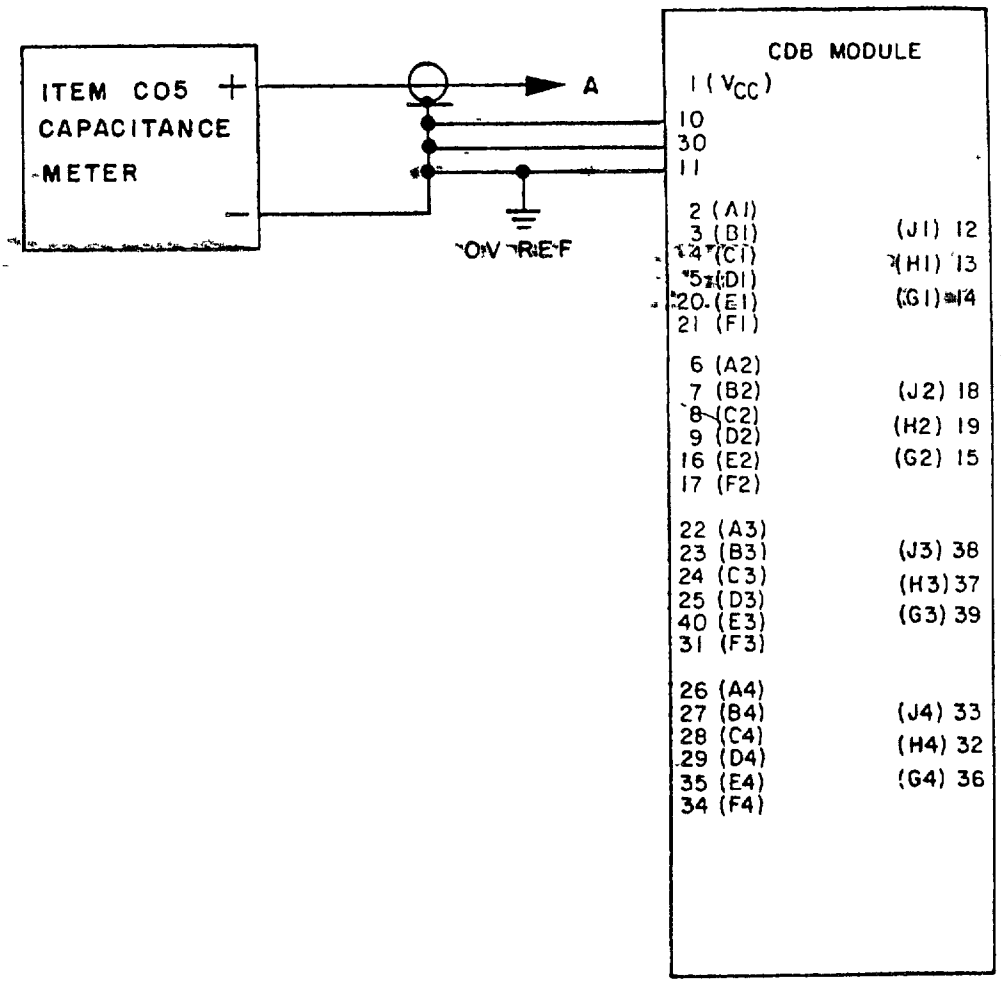
FIGURE 16. Power supply output current (low-level and high-level) test circuit for table XI.

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TABLE XII. Input capacitance test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
S1	A1 Input capacitance	2	2	16	pF	17
S2	B1	3				
S3	C1	4				
S4	D1	5				
S5	E1	20				
S6	A2	6				
S7	B2	7				
S8	C2	8				
S9	D2	9				
S10	E2	16				
S11	A3	22				
S12	B3	23				
S13	C3	24				
S14	D3	25				
S15	E3	40				
S16	A4	26				
S17	B4	27				
S18	C4	28				
S19	D4	29				
S20	E4	35				

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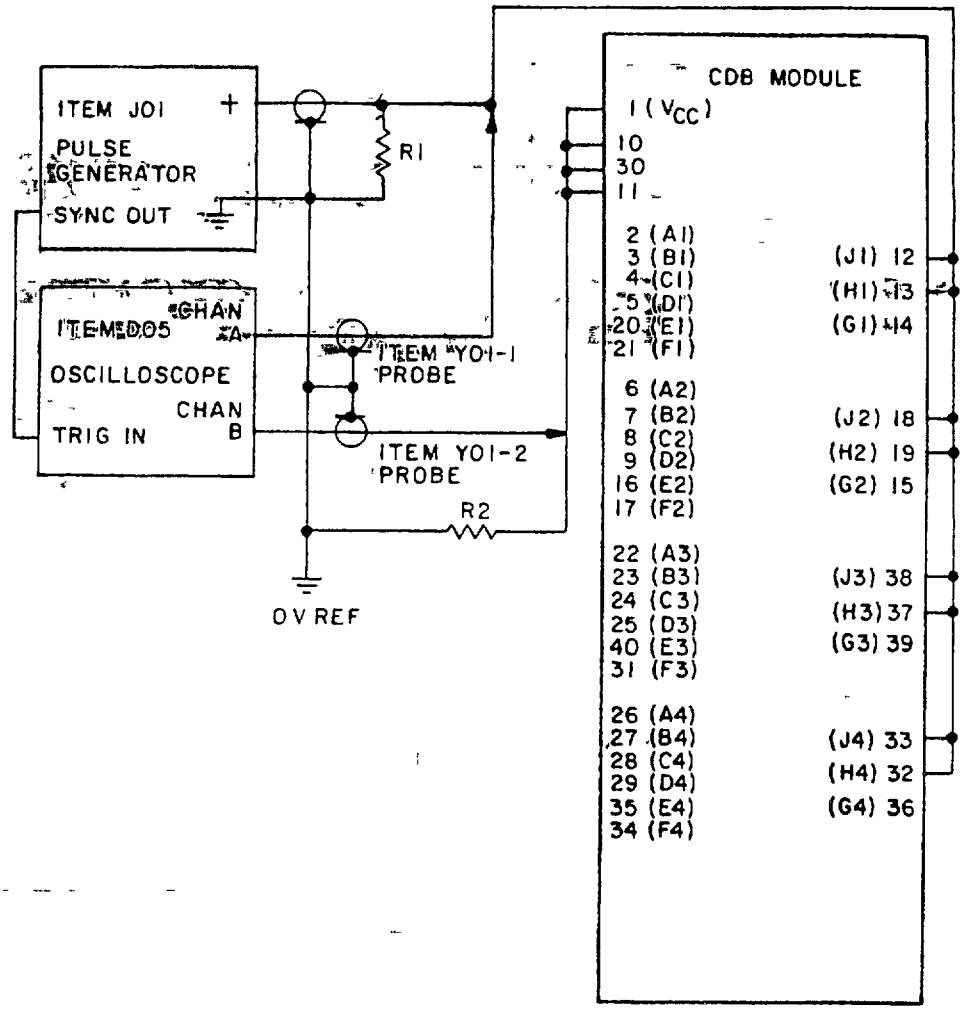
* FIGURE 17. Input capacitance test circuit for table XII.

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* TABLE XIII. Driver output isolation test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
1	Driver output isolation	0	700	mV	18,19

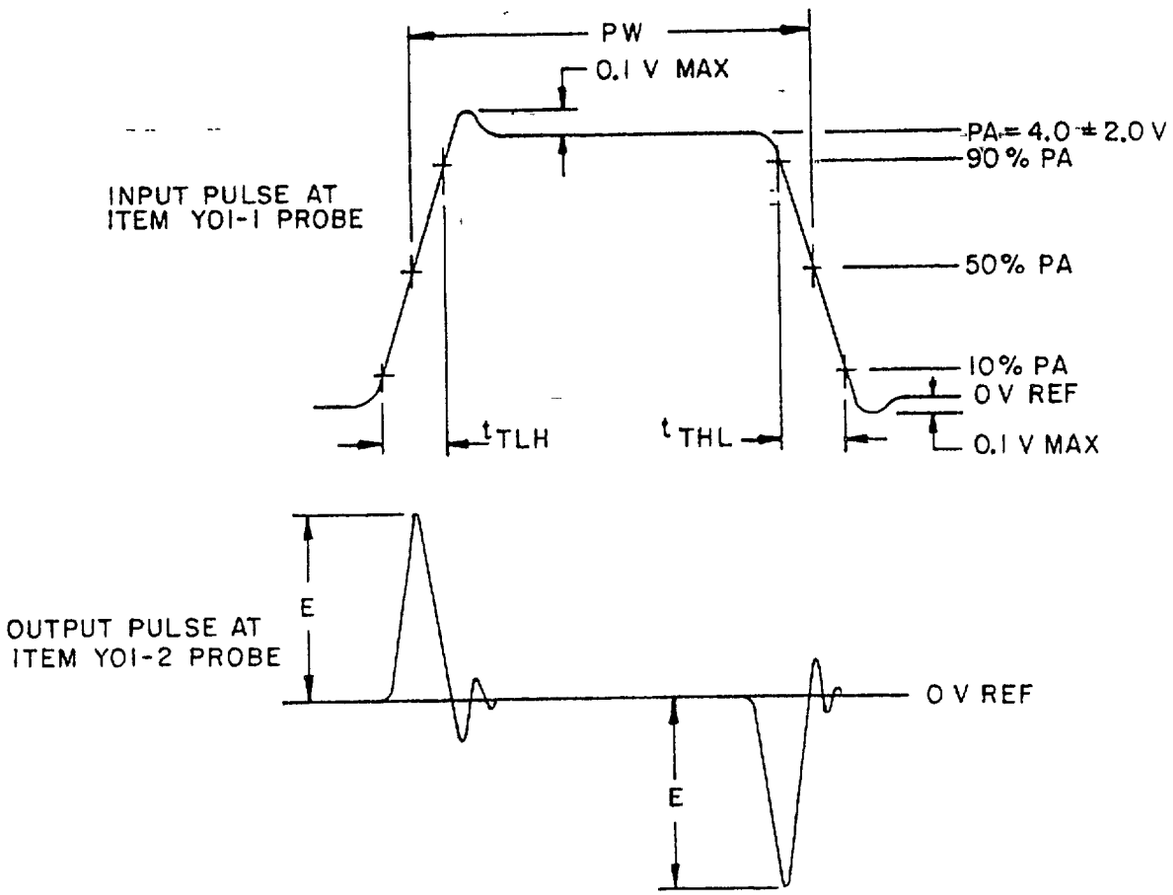
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NOTES:
1. R1 = 50Ω ±1%, 1 W.
2. R2 = 10Ω ±1%, 1/4 W.

FIGURE 18. Driver output isolation test circuit for table XIII.

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NOTES:
1. Input pulse parameters:
   PW = 100 ns ±10%
   $t_{TLH} = t_{THL}$ = 8.0 ±2.0 ns.
   PRR = 1 MHz ±10%.
2. Voltage E shall not exceed ±700 mV.

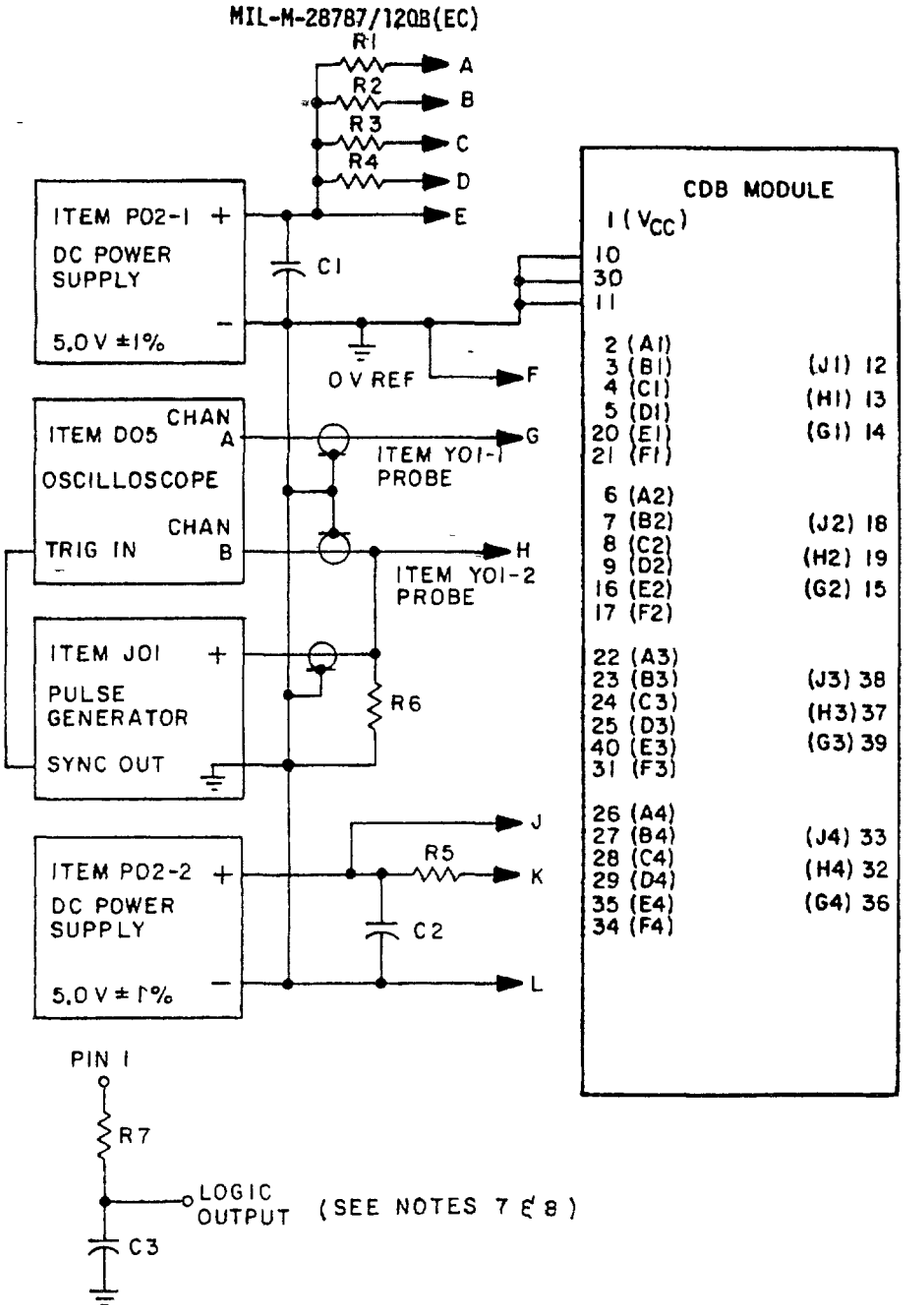
* FIGURE 19. Driver output isolation test circuit waveforms for table XIII.

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* TABLE XIV Logic and driver output crosstalk (low-level and high-level) test.

Test condition	Parameter	Connect											Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	J to pin	K to pin	L to pin	Min	Max		
U1	Logic output crosstalk, low level	13	19	37	32	Open	12,18, 38,33	14	6,22, 26	1	Open	Open	0	800	mV	20,21
U2								15	2,22, 26							
U3								39	2,6, 26							
U4								36	2,6, 22							
U5	Logic output crosstalk, high level							14	6,22, 26			2				
U6								15	2,22, 26			6				
U7								39	2,6, 26			22				
U8								36	2,6, 22			26				
V1	Driver output crosstalk, low level	Open				1	18,38, 33	13	6,22, 26	Open	13	12		400		20;22
V2		13	Open				12,38, 33	19	2,22, 26		19	18				
V3		13	19	Open			12,18, 33	37	2,6, 26		37	38				
V4		13	19	37	Open		12,18, 38	32	2,6, 22		32	33				
V5	Driver output crosstalk, high level	Open	19	37	32		18,38, 33	13	6,22, 26		13	2,12		600		
V6		13	Open	37	32		12,38, 33	19	2,22, 26		19	6,18				
V7		13	19	Open	32		12,18, 33	37	2,6, 26		37	22;38				
V8		13	19	37	Open		12,18, 38	32	2,6, 22		32	26,33				

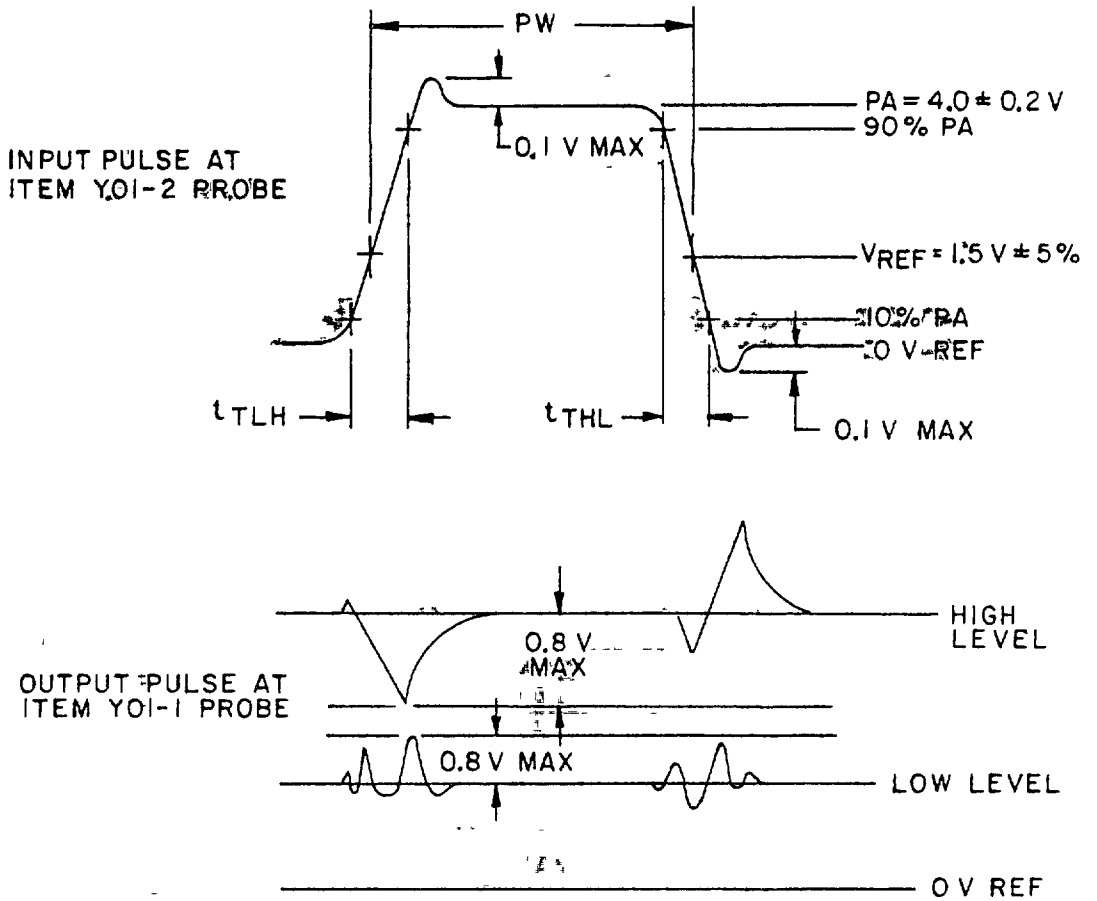
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NOTES.
1. R1-R5 = 46.4Ω ±1%, 3/4 W.
2. R6 = 50Ω ±1%, 1 W.
3. R7 = 464Ω ±1%, 1/8 W.
4. C1 = C2 = 1 µF ±10% mylar capacitor in parallel with a 330 µF ±10% tantalum capacitor. The lead length from the body of the capacitor to pin 10 shall not exceed 0.75 inches (19.0 mm).
5. C3 = 30 ±5 pF.
6. With the module removed from the test connector, measure the capacitance between each test connector pin (14, 15, 39, 36) and the ground pins (10, 11, 30). The measured values at 1 MHz ±10% shall be 30 ±5 pF. For these measurements, disconnect R7 from the respective output pin. The resistor should be disconnected as close to the body as practical. Reconnect the resistors prior to using the test connector for module testing.
7. For test conditions U1-U8, the separate load given is to be added to each pin (14, 15, 39, 36) of the logic outputs.
8. The loop length from pin 1 to each output, including the length of the body of the resistor, shall not exceed 4.5 inches (114.3 mm).

* FIGURE 20. Logic and driver output crosstalk (low-level and high-level) test circuit for table XIV.

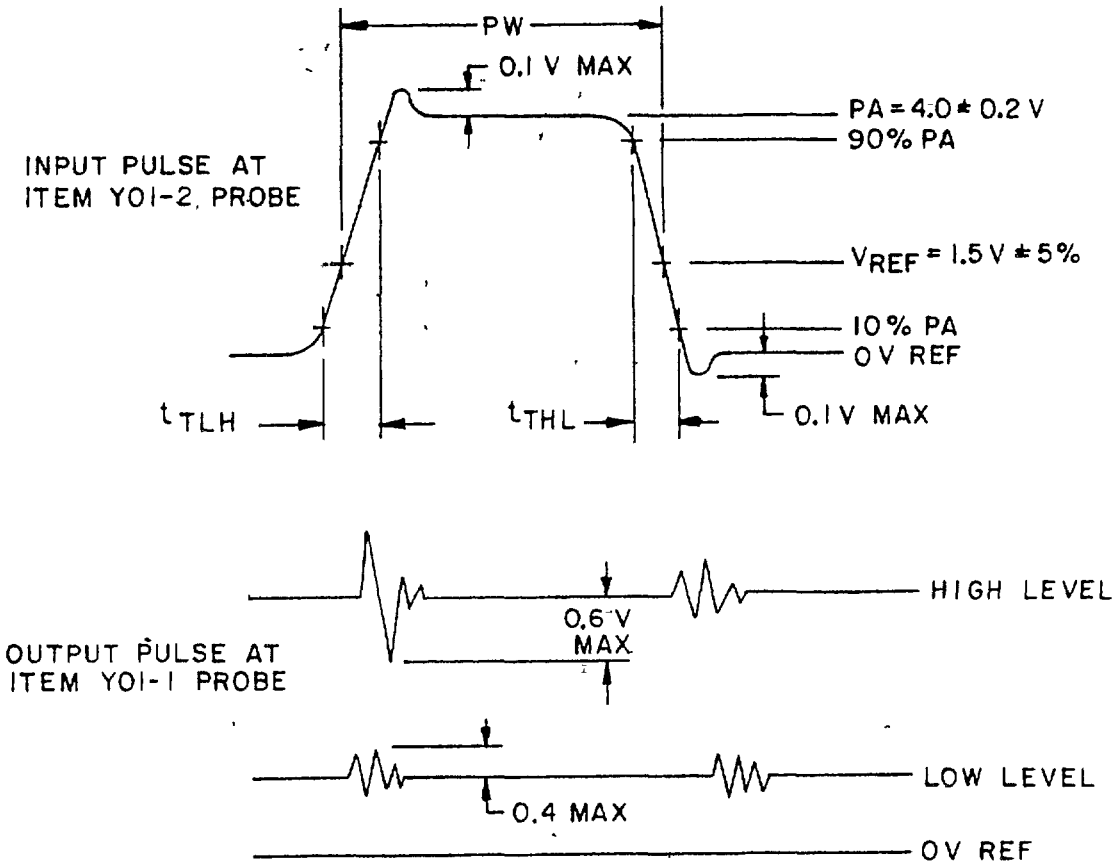
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NOTE: Input pulse parameters:
PW = 1.0 ±0.1 µs.
$t_{TLH} = t_{THL}$ = 10 ±1 ns.
PRR = 100 kHz ±10%.

*FIGURE 21. Logic output crosstalk (low-level and high-level) test circuit waveforms for table XIV.

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NOTE: Input pulse parameters:
PW = 1.0 ±0.1 µs.
$t_{TLH} = t_{THL}$ = 10 ±1 ns.
PRR = 100 kHz ±10%.

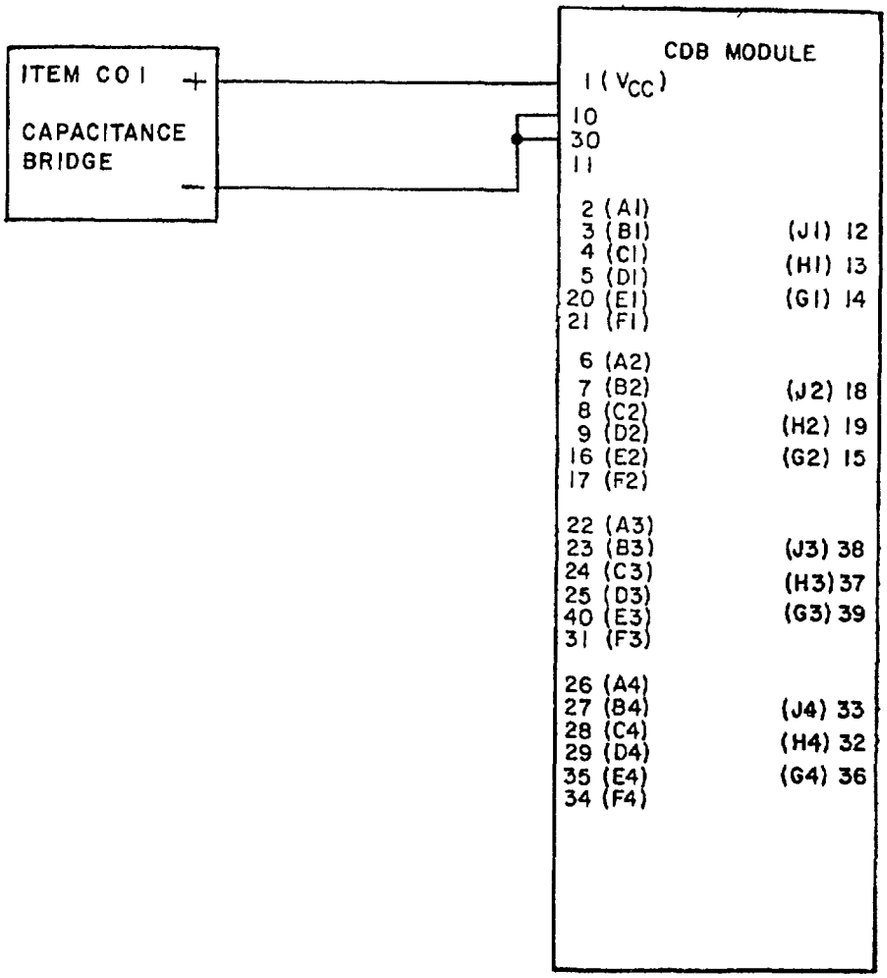
FIGURE 22. Driver output crosstalk (low-level and high-level) test circuit waveforms for table XIV.

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TABLE XV. Power supply filter capacitance test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
W	Power supply filter capacitance	1.6	3.0	µF	23

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connectors or wires to any of the module pins other than pins 10 and 30 and the pin under test.
3. Item C01 setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

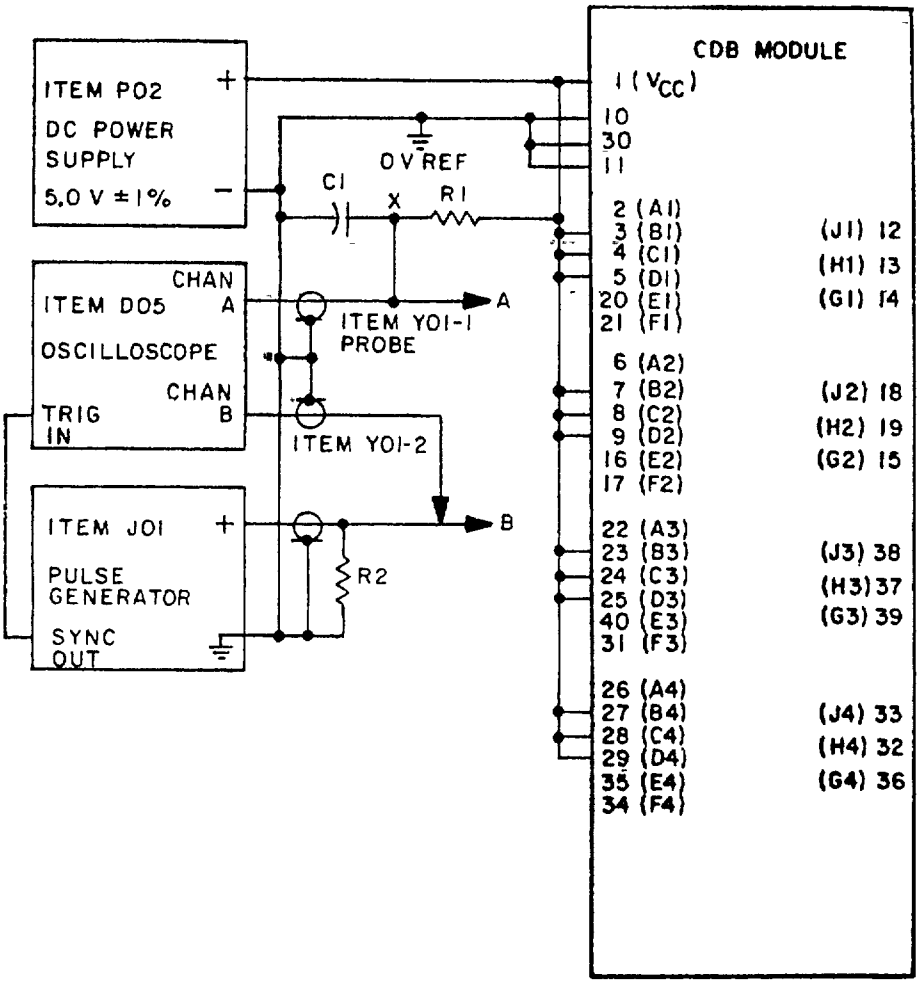
* FIGURE 23. Power supply filter capacitance test circuit for table XV.

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* TABLE XVI. Logic output voltage (high-level and low-level), transition time (t_{THL} and t_{TLH}), and propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition		Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
X1	G1	Logic output voltage, high level	14	2	3.8	5.5	V	24,25
X2	G2		15	6				
X3	G3		39	22				
X4	G4		36	26				
X5	G1	Logic output voltage, low level	14	2	0	400	mV	
X6	G2		15	6				
X7	G3		39	22				
X8	G4		36	26				
Y1	G1	Transition time (t_{THL})	14	2	10	40	ns	
Y2	G2		15	6				
Y3	G3		39	22				
Y4	G4		36	26				
Y5	G1	Transition time (t_{TLH})	14	2	110	450		
Y6	G2		15	6				
Y7	G3		39	22				
Y8	G4		36	26				
Z1	G1	Propagation delay time (t_{PLH})	14	2	25	100		
Z2	G2		15	6				
Z3	G3		39	22				
Z4	G4		36	26				
Z5	G1	Propagation delay time (t_{PHL})	14	2	3	75		
Z6	G2		15	6				
Z7	G3		39	22				
Z8	G4		36	26				

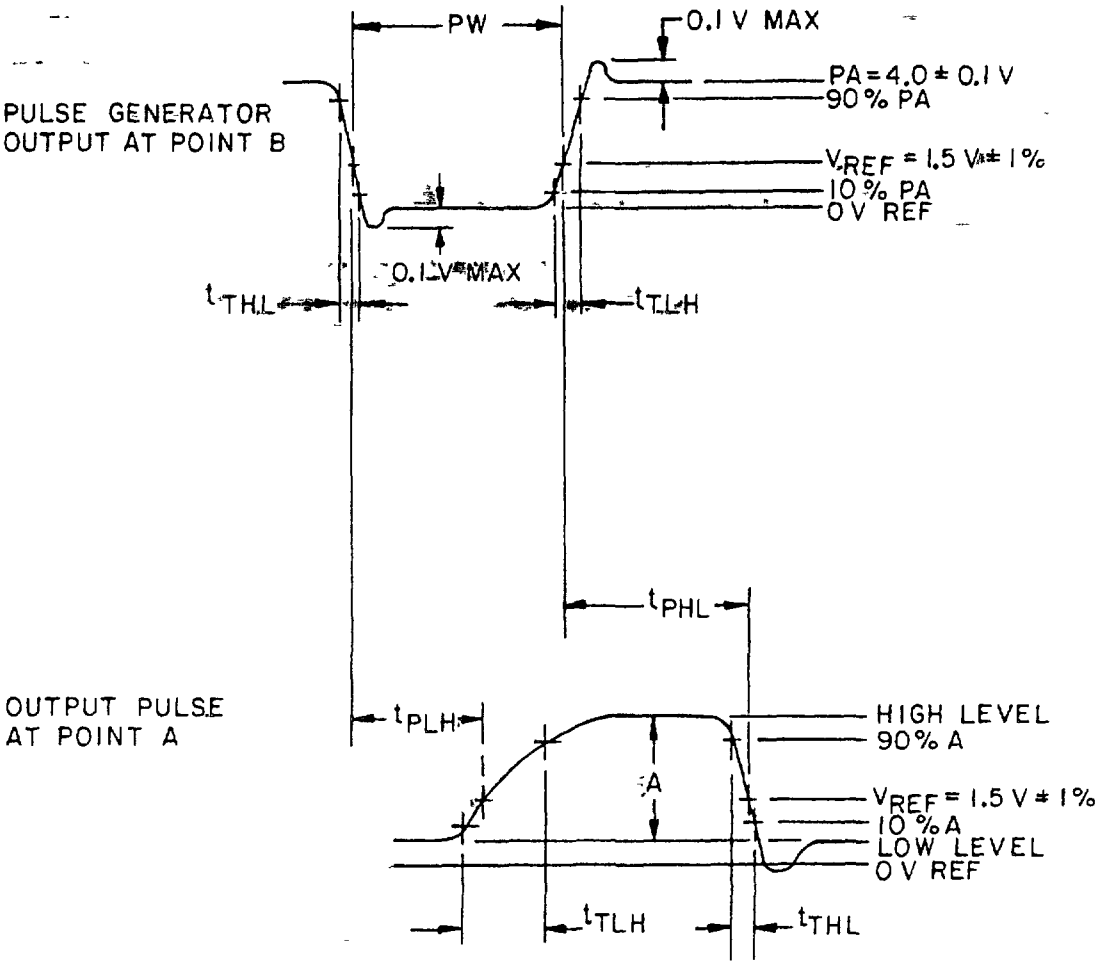
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NOTES:
1. R1 = 3 83 kΩ ±1%, 1/8 W.
2. R2 = 50Ω ±1%, 1 W.
3. Capacitor C1 shall be 30 ±5 pF when measured at 1 MHz ±10% and shall include capacitance due to wiring and oscilloscope probe. When measuring capacitor C1, the power supply must be disconnected only at the positive terminal.
4. The wire between each circuit tested and point B shall be 4 inches (101.6 mm) maximum.
5. The wire between point X and connection on module shall be 4 inches (101.6 mm) maximum.

FIGURE 24. Logic output voltage (high-level and low-level), transition time (t_{THL} and t_{TLH}), and propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XVI.

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NOTE: Input pulse parameters:
PW = 5 µs ±10%.
$t_{TLH} = t_{THL}$ = 20 ±2 ns.
PRR = 100 kHz ±10%.

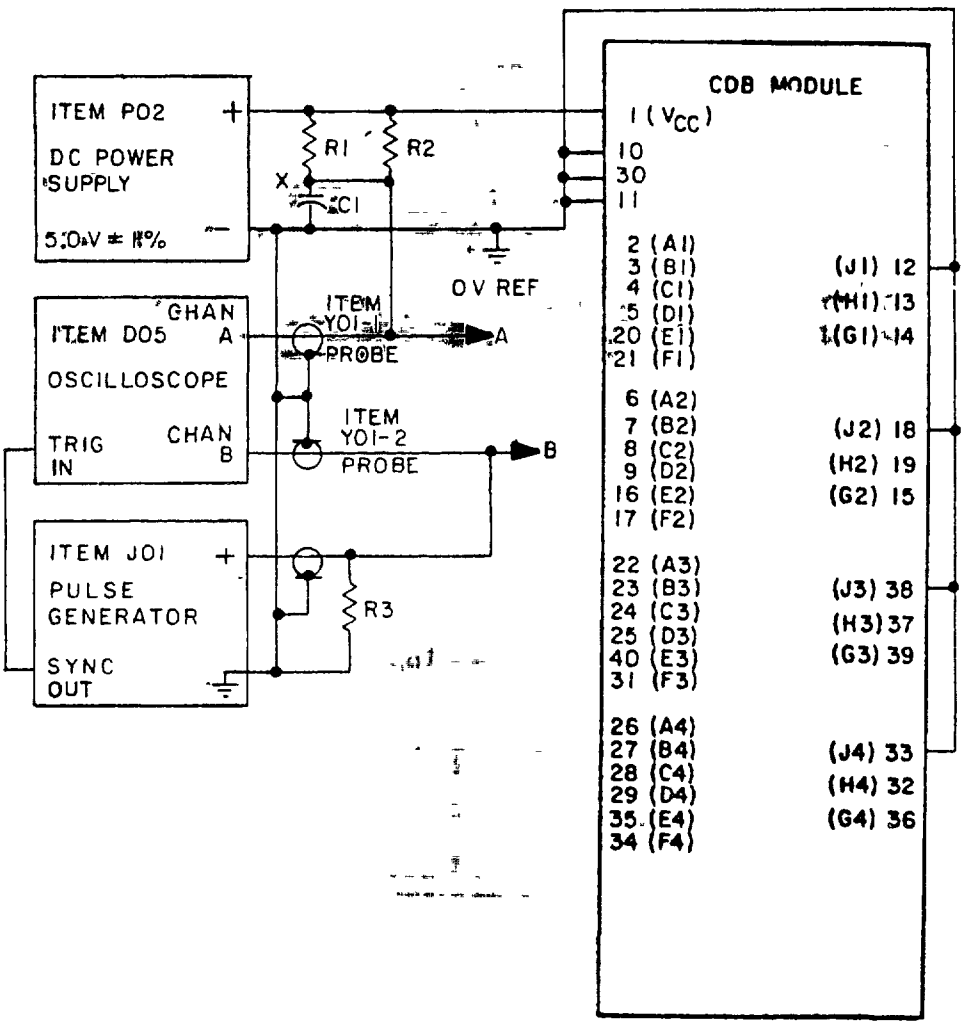
FIGURE 25. Logic output voltage (high-level and low-level), transition time (t_{THL} and t_{TLH}), and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XVI.

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* TABLE XVII. Driver output (high-level and low-level), transition time(t_{THL} and t_{TLH}), and propagation delay time (t_{PLH} and t_{PHL}) condition 1 test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
AA1	H1 Driver output (high level), condition 1	13	2,3,4,5	4.5	5.5	V	26,27
AA2	H2	19	6,7,8,9				
AA3	H3	37	22,23,24,25				
AA4	H4	32	26,27,28,29				
AA5	H1 Driver output (low level), condition 1	13	2,3,4,5	0	430	mV	
AA6	H2	19	6,7,8,9				
AA7	H3	37	22,23,24,25				
AA8	H4	32	26,27,28,29				
BB1	H1 Driver output transition time (t_{THL}), condition 1	13	2,3,4,5	8	100	ns	
BB2	H2	19	6,7,8,9				
BB3	H3	37	22,23,24,25				
BB4	H4	32	26,27,28,29				
BB5	H1 Driver output transition time (t_{TLH}), condition 1	13	2,3,4,5		150		
BB6	H2	19	6,7,8,9				
BB7	H3	37	22,23,24,25				
BB8	H4	32	26,27,28,29				
CC1	H1 Driver output propagation delay time (t_{PLH}), condition 1	13	2,3,4,5	40	275		
CC2	H2	19	6,7,8,9				
CC3	H3	37	22,23,24,25				
CC4	H4	32	26,27,28,29				
CC5	H1 Driver output propagation delay time (t_{PHL}), condition 1	13	2,3,4,5	20	175		
CC6	H2	19	6,7,8,9				
CC7	H3	37	22,23,24,25				
CC8	H4	32	26,27,28,29				

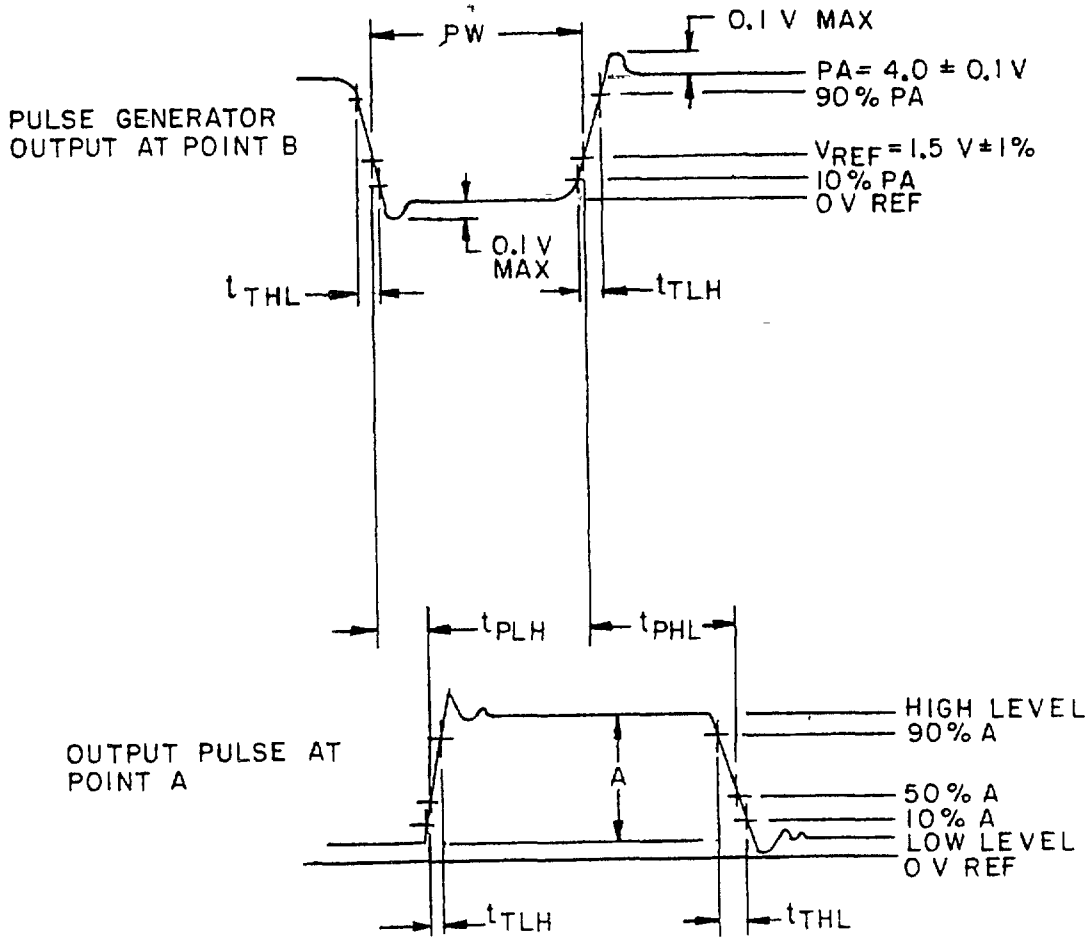
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NOTES:
1. R1 = R2 = 46.4Ω ±1%, 3/4 W.
2. R3 = 50Ω ±1%, 1 W.
3. Capacitor C1 shall be 100 pF ±10% when measured at 1 MHz ±10% and shall include capacitance due to wiring and oscilloscope probe.
4. The wire between each circuit tested and point B shall be 4 inches (101.6 mm) maximum.
5. The wire between point X and connection on module shall be 4 inches (101.6 mm) maximum.
6. The wiring capacitance at pins 21, 17, 31, and 34 shall be 10 pF or less measured at 1 MHz ±10%.

FIGURE 26. Driver output (high-level and low-level), transition time (t_{THL} and t_{TLH}), and propagation delay time (t_{PLH} and t_{PHL}) condition 1 test circuit for table XVII.

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NOTE: Input pulse parameters:
PW = 5 µs ±10%.
$t_{TLH} = t_{THL}$ = 20 ±2 ns.
PRR = 100 kHz ±10%.

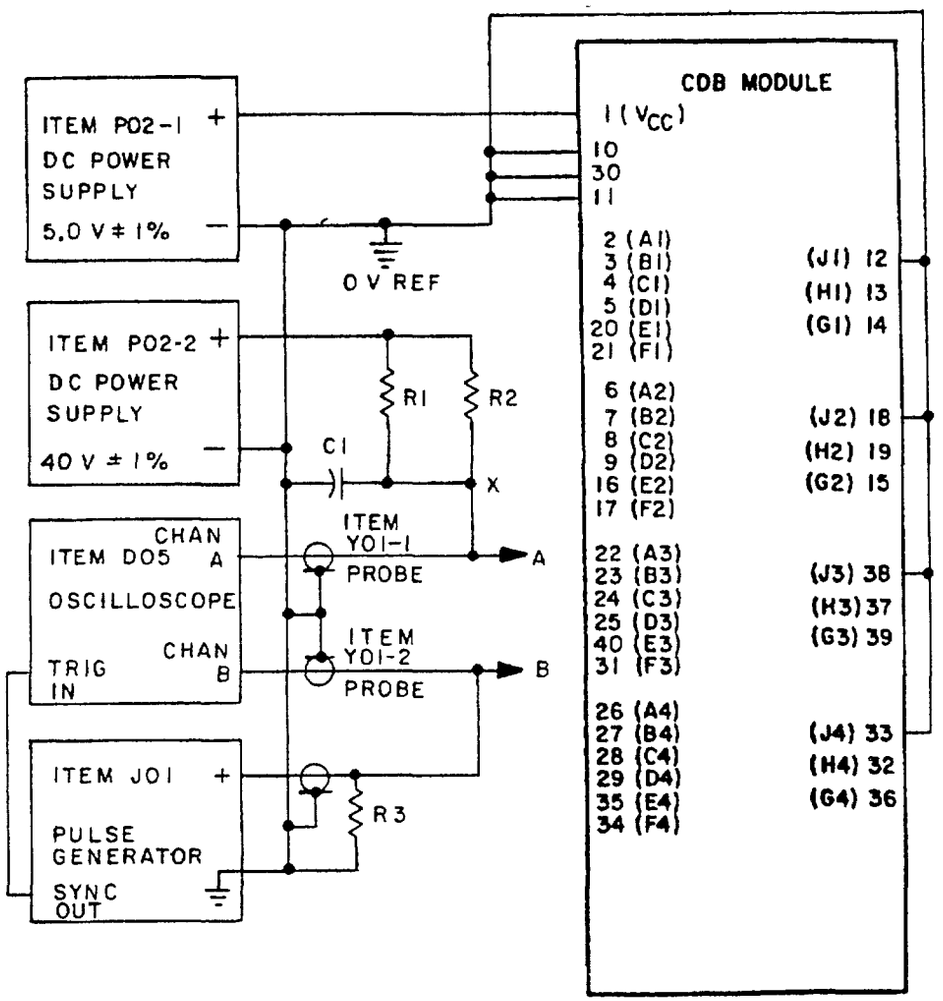
FIGURE 27. Driver output (high-level and low-level), transition time (t_{THL} and t_{TLH}), and propagation delay time (t_{PLH} and t_{PHL}) condition 1 test circuit waveforms for table XVII.

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* TABLE XVIII Driver output (high-level and low-level), transition time(t_{THL} and t_{TLH}), and propagation delay time (t_{PLH} and t_{PHL}) condition 2 test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
DD1	H1 Driver output (high level), condition 2	13	2,3,4,5	36	44	V	28,29
DD2	H2	19	6,7,8,9				
DD3	H3	37	22,23,24,25				
DD4	H4	32	26,27,28,29				
DD5	H1 Driver output (low level), condition 2	13	2,3,4,5	0	450	mV	
DD6	H2	19	6,7,8,9				
DD7	H3	37	22,23,24,25				
DD8	H4	32	26,27,28,29				
EE1	H1 Driver output transition time (t_{THL}), condition 2	13	2,3,4,5	15	130	ns	
EE2	H2	19	6,7,8,9				
EE3	H3	37	22,23,24,25				
EE4	H4	32	26,27,28,29				
EE5	H1 Driver output transition time (t_{TLH}), condition 2	13	2,3,4,5	1.0	3.0	µs	
EE6	H2	19	6,7,8,9				
EE7	H3	37	22,23,24,25				
EE8	H4	32	26,27,28,29				
FF1	H1 Driver output propagation delay time (t_{PLH}), condition 2	13	2,3,4,5	200	900	ns	
FF2	H2	19	6,7,8,9				
FF3	H3	37	22,23,24,25				
FF4	H4	32	26,27,28,29				
FF5	H1 Driver output propagation delay time (t_{PHL}), condition 2	13	2,3,4,5	30	200		
FF6	H2	19	6,7,8,9				
FF7	H3	37	22,23,24,25				
FF8	H4	32	26,27,28,29				

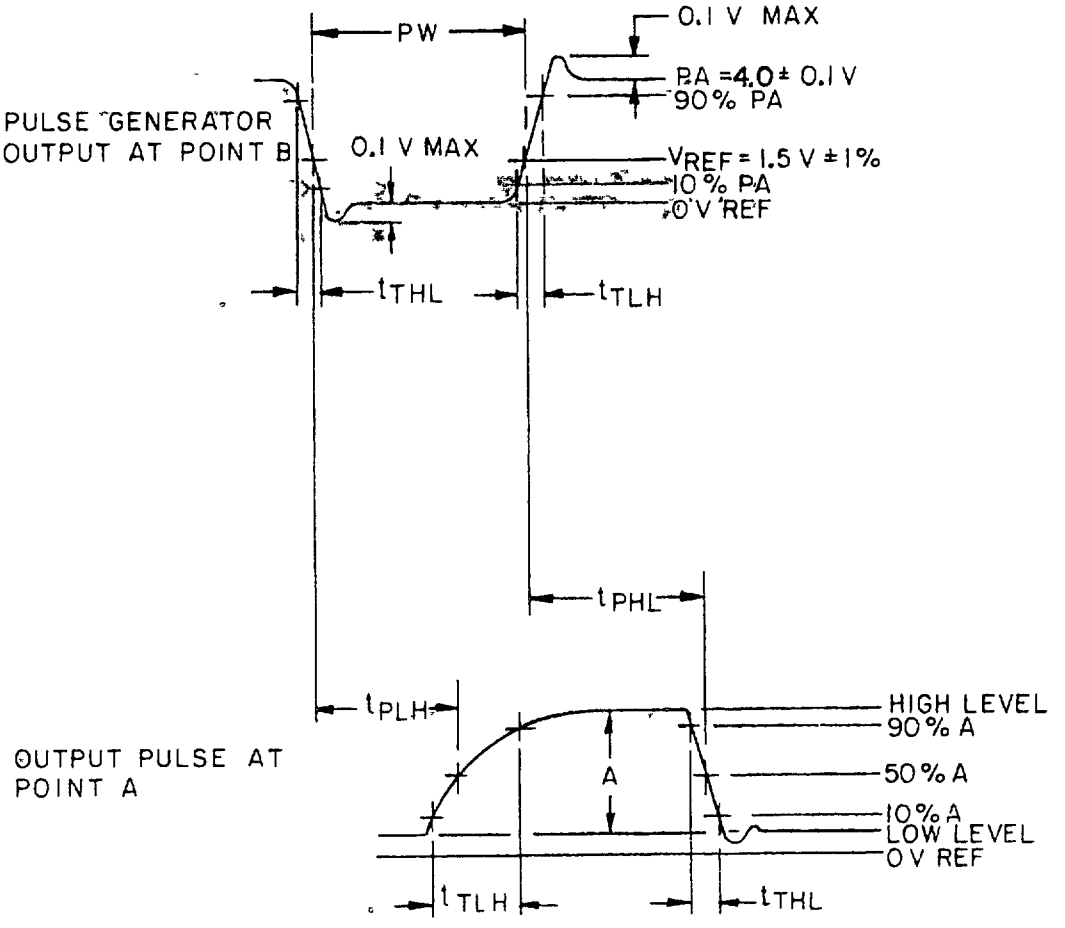
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NOTES
1. R1 = R2 = 2.15 kΩ ±1%, 3/4 W.
2. R3 = 50Ω ±1%, 1 W.
3. C1 = 680 pF ±10% at 1 kHz ±10%.
4. The wire between each circuit tested and point B shall be 4 inches (101.6 mm) maximum.
5. The wire between point X and connection to module shall be 4 inches (101.6 mm) maximum.
6. The wiring capacitance at pins 21, 17, 31, and 34 shall each be 10 pF or less measured at 1 MHz ±10%.

FIGURE 28. Driver output (high-level and low-level), transition time (t_{THL} and t_{TLH}), and propagation delay time (t_{PLH} and t_{PHL}) condition 2 test circuit for table XVIII.

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NOTE. Input pulse parameters:
PW = 5 µs ±10%.
$t_{TLH} = t_{THL}$ = 20 ±2 ns.
PRR = 100 kHz ±10%.

FIGURE 29. Driver output (high-level and low-level), transition time (t_{THL} and t_{TLH}), and propagation delay time (t_{PLH} and t_{PHL}) condition 2 test circuit waveforms for table XVIII.

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